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ORIGINAL PAPER

Piezoelectricity and flexoelectricity in biological cells: the role of cell structure and organelles

Akepogu Venkateshwarlu[1] · Akshayveer[1] · Sundeep Singh[2] · Roderick Melnik[1]

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Abstract
Living tissues experience various external forces on cells, influencing their behaviour, physiology, shape, gene expression, and destiny through interactions with their environment. Despite much research done in this area, challenges remain in our better understanding of the behaviour of the cell in response to external stimuli, including the arrangement, quantity, and shape of organelles within the cell. This study explores the electromechanical behaviour of biological cells, including organelles like microtubules, mitochondria, nuclei, and cell membranes. A two-dimensional bio-electromechanical model for two distinct cell structures has been developed to analyze the behavior of the biological cell to the external electrical and mechanical responses. The piezoelectric and flexoelectric effects have been included via multiphysics coupling for the biological cell. All the governing equations have been discretized and solved by the finite element method. It is found that the longitudinal stress is absent and only the transverse stress plays a crucial role when the mechanical load is imposed on the top side of the cell through compressive displacement. The impact of flexoelectricity is elucidated by introducing a new parameter called *the maximum electric potential ratio* ($V_{R,\max}$). It has been found that $V_{R,\max}$ depends upon the orientation angle and shape of the microtubules. The magnitude of $V_{R,\max}$ exhibit huge change when we change the shape and orientation of the organelles, which in some cases (boundary condition (BC)-3) can reach to three times of regular shape organelles. Further, the study reveals that the number of microtubules significantly impacts effective elastic and piezoelectric coefficients, affecting cell behavior based on structure, microtubule orientation, and mechanical stress direction. The insight obtained from the current study can assist in advancements in medical therapies such as tissue engineering and regenerative medicine.

Keywords Electromechanical coupling · Shape of the organelles · Electric potential ratio · Microtubule's orientation & · Medical therapies · Cell structure · Flexoelectricity

1 Introduction

Cell mechanics plays a vital role in the development, functioning, metabolism, and regulation of nuclear responses in cells. The primary factors that determine the mechanical properties of the cell are the elasticity of the cell, adhesion between the cell and its substrate, and membrane tension (Wang et al. 2015; Li et al. 2015; Chen et al. 2018; Tang et al. 2021; Torbati et al. 2022). The biological cell exhibits a highly intricate internal structure. The cell comprises two primary constituents: the cytoplasm and nucleus. The cytoplasm is a fluidic matrix that commonly surrounds the nucleus and is enclosed by the cellular plasma membrane (Holy 2003; Link et al. 2023).

Organelles are subcellular structures that are microscopic and are located within the cytoplasm. These structures are responsible for carrying out various functions that are

✉ Akepogu Venkateshwarlu
avenkateshwarlu@wlu.ca

Akshayveer
aakshayveer@wlu.ca

Sundeep Singh
sunsingh@upei.ca

Roderick Melnik
rmelnik@wlu.ca

[1] MS2Discovery Interdisciplinary Research Institute, Wilfrid Laurier University, 75 University Avenue West, Waterloo, ON N2L 3C5, Canada

[2] Faculty of Sustainable Design Engineering, University of Prince Edward Island, Charlottetown, PE C1A 4P3, Canada

essential for maintaining the stability of the cell's internal environment, also known as homeostasis (Basoli et al. 2018; Shahab et al. 2023). These functions include energy generation, protein synthesis and release, toxin removal, and the reception of external stimuli. Electrical stimulation therapy has garnered significant interest due to its advantages in stimulating cell proliferation and tissue regeneration, expediting the restoration of the skin barrier, and diminishing scarring (Song et al. 2023; Xu et al. 2023). Clinical evidence has demonstrated its efficacy in expediting wound healing. For instance, mast cells, found in various body parts like the skin, airways, gastrointestinal system, and mucosa, are quick to respond to both internal and external stimuli (Yi et al. 2018). Neutrophils are constantly exposed to hemodynamic forces, including shear stress, which regulates their biological activities like crawling and NETosis (Zhu et al. 2020; Zeng et al. 2023). The organelles undergo alterations by changing their shape, quantity, and spatial arrangement inside the cell, which affects their structure and organization (Zhang and Assouline 2007; Lin et al. 2012; Hoyer-Fender 2013; Park 2023). Cellular dynamics are directly impacted by processes such as division, merging, morphological changes, transportation, and external stimuli (Zhu et al. 2020; Fenton et al. 2021; Liang et al. 2023-07). For instance, mitochondria are attached to the cytoskeleton at precise sites within the cell. Neurons rely on mitochondrial anchoring to ensure that mitochondria are solidly attached to presynaptic locations in axons, enabling them to provide the necessary energy for neurotransmission (Chada and Hollenbeck 2004; Gutnick et al. 2019; Cason and Holzbaur 2022).

The regulation of microtubule dynamics is crucial for the proper functioning and division of all eukaryotic cells. The wide range of actions is primarily facilitated by the dynamic ends of microtubules, which can attach to different targets within the cell, produce mechanical forces, and interact with actin microfilaments (Rodríguez-García et al. 2020). Microtubules are cellular structures that play an essential part in understanding how related proteins, mechanical forces, and microtubule-targeting medicines, such as cancer chemotherapeutics, regulate the mechanisms of microtubule function (Gudimchuk and McIntosh 2021; Gudimchuk and Alexandrova 2023). Microtubules also play a crucial role in facilitating the transportation of diverse cellular structures, such as lysosomes, mitochondria, and other organelles (Tuszynski et al. 2020; Kummer and Ban 2021). The structure of microtubules is based on dimers composed of α and β-tubulin, which exhibit an elongated, non-branching polymer pattern (Li et al. 2017). Each microtubule consists of approximately thirteen dimeric tubulin molecules arranged in a circular configuration (Setayandeh and Lohrasebi 2016). The length of the microtubule can be modified through the addition or removal of dimers. The microtubules, which have rigid structures found within cells, can cause significant strain fluctuations due to their small size and interaction with a flexible cellular matrix. Further, they provide stiffness to the structures, and their number depends on cell type, structural characteristics, and the primary function (Ishikawa 2017). Among the many cell types that may be found in a wide variety of organisms, ranging from protists to mammals, cilia and flagella are two structures that are particularly notable and widely recognized. Protists have the ability to arrange microtubules in various diverse formations, which results in the construction of structures that are both unique and comprehensive de Souza and Attias (2010). Some motile cilia contain different numbers of microtubules, such as singlets in apicomplexans, fungi, and Caenorhabditis elegans (Mollaret and Justine 1997; Bezler et al. 2022), symmetrical basal bodies with nine microtubules, and complex organelles like motile cilium with 9 + 2 structures (Zhang and Assouline 2007; Lin et al. 2012; Hoyer-Fender 2013). Axonemes in protozoans, insects, arthropods, and nematodes have varying numbers of microtubule doublets, precisely 10, 12, 14, 16, or 18 (Roggen et al. 1966; van Deurs 1973; Dallai et al. 1992; Spreng et al. 2019; Ferreira et al. 2023). External stimuli increase actin stress fibres in cells exposed to shear stress or geometric patterns, resulting in an increase in the cytoskeletal network (Guolla et al. 2012).

Many computational studies (Liu et al. 2012; Yang et al. 2015; Joshan and Santapuri 2022; Thai et al. 2022; Wang and Xue 2023) have focused on the mechanical coupling between strain and stress in nanowires and plates. It has been reported that a significant proportion of biological components, such as microtubules, bones, and collagen, exhibit piezoelectric properties. For instance, during walking, the human tibia produces a piezoelectric potential of approximately 300 μV (Iwao 1977). Piezoelectric materials possess the capacity to revolutionize treatment methodologies by enabling precise, targeted medication release, reducing toxicity, identifying viral infection, and enhancing effectiveness in cancer therapy (Naqvi et al. 2020; Li et al. 2022; Jiang et al. 2023; Liang et al. 2023-07; Bin et al. 2024; Mathey-Andrews 2024; Xu et al. 2024). Robotic cell manipulation technologies exploit the principle of piezoelectricity for the detection and manipulation of the cells in the body (Hu et al. 2022). Therefore, it is important to conduct a thorough investigation into the mechanics of the cell and its organelles in order to gain a better understanding of their piezoelectric properties. The discovery of piezoelectricity has a long history, starting with the Curie brothers, who, in 1880, observed that some crystals display both positive and negative charges under pressure, which disappear once the pressure is released. This resulted in their discovery of piezoelectricity. Later, scientists used the term "piezoelectricity" to describe the direct relationship between mechanical stress and electricity (Fukada 1955; Fukada and Yasuda 1957, 1964; Fukada 1968, 1983, 2000; Chae et al. 2018). Since then, piezoelectricity,understood as the direct relationship between electrical and mechanical energy, is widely used in various applications, including artificial muscles, wearable sensors, advanced microscopes, flexible actuators, energy harvesting devices, and minimally invasive surgery (Shamos and Lavine 1967; Anderson and Eriksson 1970; Fukada 1983; Fletcher and Mullins 2010; He et al. 2017; Lun et al. 2020; Atef and El-Dhaba 2022; Rout and Kapuria 2023). On the other hand, flexoelectricity is a bidirectional linear interaction between the electric field and strain gradients, unlike piezoelectricity, which occurs due to electric polarization and strain. It is characterized by strain gradients or inhomogeneous strains, while uniform strains cause piezoelectricity (Petrov 2002; Deng et al. 2014; Lun et al. 2020; Atef and El-Dhaba 2022; Xia et al. 2024). While non-centrosymmetric materials in physical and biological systems exhibit linear piezoelectricity, which is the coupling of a single strain component with a single electric field component (He et al. 2017; Rout and Kapuria 2023), flexoelectricity can occur even in materials with centrosymmetry, especially when inhomogeneities at micro length scales lead to significant strain gradients and electric field formation (Petrov 2007; Deng et al. 2017; Abdollahi et al. 2019). Despite their minimal effects, often at larger scales, flexoelectric contributions to electromechanical coupling are likely to be substantial at smaller scales (Moura and Erturk 2017; Krichen and Sharma 2016; Wang et al. 2019; Thai et al. 2022; Li and Du 2024).

Given these coupled effects, it becomes apparent that further research is necessary to fully understand how living cells behave, especially in the context of their variable organelle shapes. The structure and orientation of the organelles are essential for comprehending the dynamics of the cell. Moreover, the quantity of microtubules within a cell plays a crucial role in cell dynamics. These aspects have not yet been thoroughly investigated (Spreng et al. 2019; Ferreira et al. 2023). Recent experimental studies (Zhang et al. 2023; Kobori et al. 2024) show that microtubules are mostly composed of α- and β-tubulin dimerization, resulting in a polarized tubular structure. Cells can extend and contract at the plus end. Cell morphology may depend on microtubule reorientation. Non-centrosomal microtubules (NC-MTs) may affect cell adhesion and division, depending on cell type. However, these implications have not been explored in the context of piezoelectric and flexoelectric effects. Though significant progress has been made, much remains unknown about how individual organelles operate and behave as a whole within living cells when subjected to external stimuli like flexoelectricity and piezoelectricity (Fletcher and Mullins 2010; He et al. 2017; Lun et al. 2020; Atef and El-Dhaba 2022; Thai et al. 2022; Rout and Kapuria 2023). The significance of mechanical stimuli in regulating cell functions is becoming even more apparent, given the fact that the understanding of the immediate structural reaction of cells to mechanical stresses is still inadequate (Guolla et al. 2012; Xia et al. 2024). Therefore, the primary aim of this work is to investigate the impact of externally exerted mechanical displacement on the characteristics of individual cells, with a specific emphasis on the variable shapes and sizes of organelles.

The main highlights of the present work are that the structures of biological cells have been created in regular and irregular shapes in order to elucidate the real behaviour of the biological cell. The shape of the microtubule is created in such a way that its edges mimic the joining of the α-tubulin and β-tubulin in a real microtubule. The quantity of the microtubules is also varied because within a cell the microtubule plays an important role in cell dynamics. Furthermore, the present work has taken into consideration the influence of orientation on the piezoelectric tensor of microtubules. We now anticipate that irregularly shaped organelles will exhibit greater flexoelectric and piezoelectric effects than regularly shaped organelles. Therefore, we expect to see more effective responses from external electromechanical loads in irregular shapes compared to regular shapes. This would enable our geometry designed with irregular organelle shapes to closely mimic the structure of a real biological cell. In what follows, we carry out an extensive parametric analysis to examine the effects of piezoelectricity and flexoelectricity by varying the number of microtubules from 2 to 18 along with their shapes when the cells are subjected to the mechanical displacements. The mechanical response of the cell is elucidated by determining the effective elastic coefficients, and the contribution of piezoelectricity resulting from mechanical displacement is quantified through the effective piezoelectric coefficients.

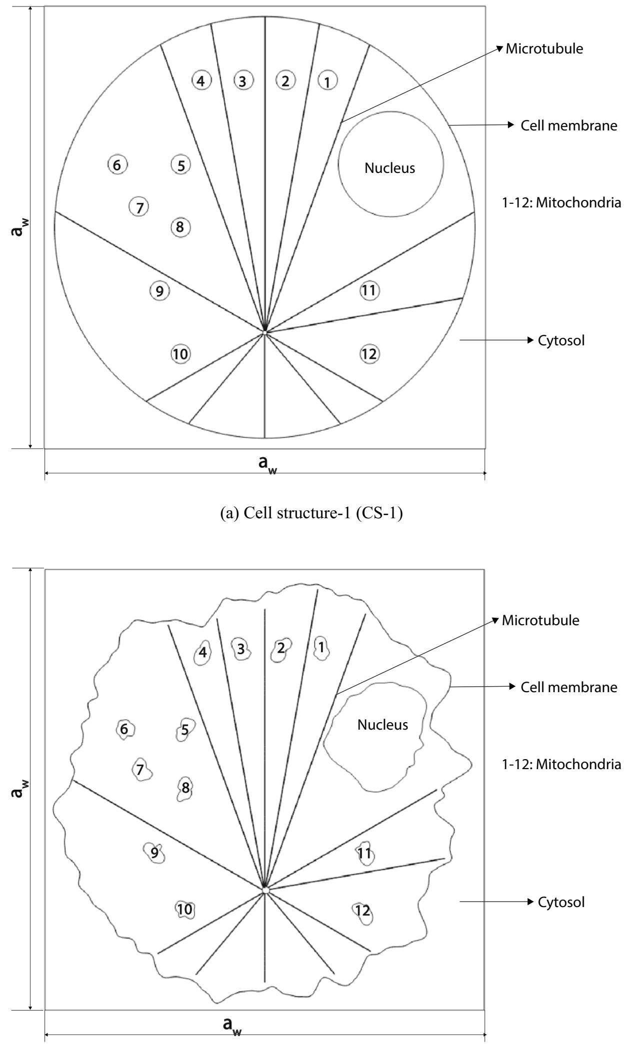
The rest of the paper has been structured as follows: Sect. 2 describes the model that governs the coupled electro-elastic behaviour of the cells, including the geometry of the cell structure and the boundary conditions. The results obtained from the present study, which include the effects of piezoelectricity and flexoelectricity on the distribution of electric potential, strain, effective elastic coefficients, and effective piezoelectric coefficients, are delineated in Sect. 3. The findings are summarized in Sect. 4.

2 Physics-based model of the biological cell

This section details the developed model of the biological cell, including material properties, mathematical framework, numerical setup, and boundary conditions necessary for problem solving. The cell has a width of $a_w = 21\,\mu\mathrm{m}$ (Connie et al. 2016; Natasha 2016; Singh et al. 2020). It contains distinct organelles with well-defined shapes in cell structure-1 in the $x_1 - x_3$ plane. However, the realistic organelles lack a specific and well-defined shape. Therefore, we have also selected arbitrary shapes for intracellular organelles, such as the nucleus, mitochondria, cytoplasm, microtubules, cell membrane, and centrosome, for cell structure-2. The arbitrary shapes of the organelles have been created using an in-house developed MATLAB code. The organelles possess unique structures that exhibit a diverse range of sizes and shapes within cellular compartments. For instance, mitochondria are portrayed as rigid, long, cylinder-shaped structures that resemble bacteria. They are highly elastic and dynamic organelles that are always undergoing shape changes (Uzman 2003; Parker et al. 2017; Fenton et al. 2021). The morphology of the nucleus (Tange et al. 2002; Norppa and Falck 2003; Goldman et al. 2004; Webster et al. 2009; Khatau et al. 2012), cell membrane (Norppa and Falck 2003; Mannella 2006; Bozelli and Epand 2020), and mitochondria (Kalt 1975; Paumard et al. 2002; Mannella 2006; Velours et al. 2009) exhibits distinct characteristics in the cell structures, as shown in Fig. 1. The shape of the cell model has been created based on the experimental studies (Vaziri and Gopinath 2008; Connie et al. 2016; Natasha 2016; Moujaber and Stochaj 2020; Parker et al. 2017) as depicted in Figs. 1(a), (b). Similarly, the shapes of organelles such as the nucleus (Tange et al. 2002; Norppa and Falck 2003; Goldman et al. 2004; Webster et al. 2009; Khatau et al. 2012), cell membrane (Norppa and Falck 2003; Mannella 2006; Bozelli and Epand 2020), and mitochondria (Kalt 1975; Paumard et al. 2002; Mannella 2006; Velours et al. 2009), are modelled based on experimental and other computational studies. The dimensions of the organelles have been taken as follows (Ofek et al. 2009): nucleus, 5 µm (Connie et al. 2016); mitochondria, 0.92 µm (Connie et al. 2016); microtubule, 25 nm (diameter), length 5–15 µm (Cooper 2000; Moujaber and Stochaj 2020; Parker et al. 2017); cell membrane, 20 µm (diameter), 7 µm (Connie et al. 2016; Natasha 2016).

Both cell structures are presumed to possess an equivalent quantity of organelles, comprising 12 mitochondria, 18 microtubules, 1 nucleus, 1 cytosol, and 1 cell membrane, for the purpose of conducting a comparison analysis. Furthermore, it has been considered that all intercellular organelles have identical surface areas compared to their counterparts in the cell structures. The lengths of the microtubules are varied, and their inward ends have been connected to the centrosome. The cell membrane has been conceptualized as a structural barrier that delineates the intracellular compartment of the cell from the extracellular matrix environment (Gudimchuk and McIntosh 2021).

The material properties pertaining to the organelles that have been utilized in the current study are outlined in Table 1. The assumptions that have been postulated pertaining to the present study are as follows: the piezoelectric and flexoelectric coefficients of the organelles other than the microtubules are assumed to be zero because the main emphasis is on the



(a) Cell structure-1 (CS-1)

(b) Cell structure-2 (CS-2)

Fig. 1 Schematic representation of the structure of the two-dimensional biological cell, including various organelles such as the cell membrane, nucleus, mitochondria, and microtubules, with **(a)** idealized shapes and **(b)** shapes that closely resemble the experimental studies

piezoelectric and flexoelectric coefficients which is related to microtubules (Brown and Tuszynski 1999). The presence of transversely isotropic piezoelectric material along the x_3-axis, the electromechanical phenomena can be investigated using either the x_1-x_3 or x_2-x_3 planes (Krishnaswamy et al. 2019a). The piezoelectric coefficients of the microtubules are equivalent to those of collagen (Katti and Katti 2017). The behavior of the biological cell is assumed to be linearly elastic material (Katti and Katti 2017; Kollmannsberger and Fabry 2011; Garcia and Garcia 2018b).

The fully coupled electromechanical model, based on the extended linear theory of continuum mechanics for dielectric continua, has been used to analyze the

Table 1 Physical properties of the organelles considered in the current study

S.No.	Parameter	Elastic modulus (E)	Poisson's ratio (v)	Relative permittivity (dielectric constant)
1	Nucleus	1 kPa (Katti and Katti 2017)	0.3 (Katti and Katti 2017)	80 (Gowrishankar et al. 2006; Tiwari et al. 2009)
2	Mitochondria	50.3 kPa (Li et al. 2015)	0.3 (Singh et al. 2020)	80 (Gowrishankar et al. 2006; Tiwari et al. 2009)
3	Microtubule	2 GPa (Barreto et al. 2013)	0.3	40 (Li et al. 2015)
		1.9 GPa (Katti and Katti 2017)	0.3 (Pampaloni and Florin 2008; Barreto et al. 2013; Katti and Katti 2017)	
4	Cell membrane	1.8 kPa (Katti and Katti 2017)	0.3 (Gowrishankar et al. 2006; Tiwari et al. 2009)	3 (Gowrishankar et al. 2006; Tiwari et al. 2009)
5	Cytoplasm	0.25 kPa (Ingber et al. 2000; Barreto et al. 2013; Katti and Katti 2017)	0.49 (Ingber et al. 2000; Gowrishankar et al. 2006; Tiwari et al. 2009; Barreto et al. 2013)	80 (Gowrishankar et al. 2006; Tiwari et al. 2009)
6	Actin cortex	2 kPa (Stricker et al. 2010; Barreto et al. 2013)	0.3 (Stricker et al. 2010; Barreto et al. 2013)	

two-dimensional biological cell domain in a steady-state regime. Under equilibrium conditions and based on Gauss's law, the governing equations for the current two-dimensional computational domain (Sharma et al. 2010; Zubko et al. 2013; Mao and Purohit 2014) are given as:

$$\left(\sigma_{ij} - \hat{\sigma}_{ij,k}\right)_{,j} = 0, \tag{1}$$

$$D_{i,i} = 0, \tag{2}$$

where σ_{ij} are the components of the mechanical stress tensor, $D_{i,i}$ are the components of the electric displacement vector, and $\hat{\sigma}_{ijk}$ is the higher-order flexoelectric tensor. σ_{ij}, $D_{i,i}$, and $\hat{\sigma}_{ijk}$ have been calculated by their fundamental constitutive relationships based on the Gibbs free energy equation (G) as follows (Zubko et al. 2013; Ghasemi et al. 2017):

$$G = \frac{1}{2}c_{ijkl}\varepsilon_{ij}\varepsilon_{kl} - e_{kij}E_k\varepsilon_{ij} - \frac{1}{2}\epsilon_{ij}E_iE_j - \frac{1}{2}B_{klij}E_kE_l\varepsilon_{ij} - \mu_{ijkl}E_i\varepsilon_{jk,l}, \tag{3}$$

$$\sigma_{ij} = \frac{\partial G}{\partial \varepsilon_{ij}} = c_{ijkl}\varepsilon_{kl} - e_{kij}E_k, \tag{4}$$

$$\hat{\sigma}_{ijk} = \frac{\partial G}{\partial \varepsilon_{jk,l}} = -\mu_{ijkl}E_i = \mu_{lijk}E_l, \tag{5}$$

$$D_i = -\frac{\partial G}{\partial E_i} = \epsilon_{ij}E_j + e_{ijk}\varepsilon_{jk} + \mu_{ijkl}\varepsilon_{jk,l}. \tag{6}$$

In Eq. 3, $\frac{1}{2}c_{ijkl}\varepsilon_{ij}\varepsilon_{kl}$ is the elastic deformation energy, $e_{kij}E_k\varepsilon_{ij}$ is piezoelectric energy, $\frac{1}{2}\epsilon_{ij}E_iE_j$ is dielectric energy (it is purely due to electric field and electric properties), $\frac{1}{2}B_{klij}E_kE_l\varepsilon_{ij}$ is electrostrictive energy, and $\mu_{ijkl}E_i\varepsilon_{jk,l}$ is the flexoelectric energy. In the present study, the electrostrictive energy term has been neglected. Moreover, ε_{ij} are the components of the mechanical strain tensor, E_k are the components of the electric field vector, c_{ijkl} are the linear elastic coefficients in the stiffness matrix, e_{ijk} are the linear piezoelectric coefficients, and ϵ_{jk} are the dielectric permittivity coefficients, $\varepsilon_{jk,l}$ is the strain gradient and μ_{ijkl} is the fourth-order flexoelectric tensor.

By employing the two-dimensional biological cell model described above and assuming the presence of transversely isotropic piezoelectric material along the x_3-axis, the electromechanical phenomena can be investigated using either the x_1-x_3 or x_2-x_3 planes. Therefore, when considering the x_1-x_3 plane, using the Voigt notation, the constitutive relations (Eqs. 4-6) can be reduced into matrix form, as follows (Singh et al. 2020; Krishnaswamy et al. 2019a, b, c):

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{33}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}c_{11} & c_{13} & 0\\ c_{13} & c_{33} & 0\\ 0 & 0 & c_{44}\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{33}\\ 2\varepsilon_{13}\end{bmatrix} - \begin{bmatrix}0 & e_{31}\\ 0 & e_{33}\\ e_{15} & 0\end{bmatrix}\begin{bmatrix}E_1\\ E_3\end{bmatrix}, \tag{7}$$

$$\begin{bmatrix}D_1\\ D_3\end{bmatrix} = \begin{bmatrix}0 & 0 & e_{15}\\ e_{31} & e_{33} & 0\end{bmatrix}\begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{33}\\ 2\varepsilon_{13}\end{bmatrix} + \begin{bmatrix}\epsilon_{11} & 0\\ 0 & \epsilon_{33}\end{bmatrix}\begin{bmatrix}E_1\\ E_3\end{bmatrix} + \begin{bmatrix}\mu_{1111} & \mu_{1331}\\ \mu_{3113} & \mu_{3333}\end{bmatrix}\begin{bmatrix}\varepsilon_{11,1} & \varepsilon_{33,1}\\ \varepsilon_{11,3} & \varepsilon_{33,3}\end{bmatrix}. \tag{8}$$

The relationship between the strain field and displacement vector, electric potential, and electric field is mathematically expressed (Sharma et al. 2010; Zubko et al. 2013; Mao and Purohit 2014) as

$$\varepsilon_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right), \tag{9}$$

$$E_i = -V_{,i}. \tag{10}$$

Collagen and microtubule-associated tau proteins exhibit comparable crystalline homology (de Garcini et al. 1990; Zhu and Qian 2022). As a result, the generation of piezoelectric potential is comparable in both proteins when subjected to similar load conditions (Kushagra 2015). Therefore, in the present study, it has been assumed that the piezoelectric coefficients of the microtubules are equivalent to those of collagen (de Garcini et al. 1990; Zhu and Qian 2022). The hexagonal symmetry has been taken into consideration in order to determine each piezoelectric coefficient for collagen. The piezoelectric tensor can be represented using Voigt's notation as follows (Denning et al. 2017; Singh et al. 2020):

$$d_{ij} = \begin{bmatrix} 0 & 0 & 0 & d_{14} & d_{15} & 0 \\ 0 & 0 & 0 & d_{15} & -d_{14} & 0 \\ d_{31} & d_{31} & d_{33} & 0 & 0 & 0 \end{bmatrix}. \tag{11}$$

In the piezoelectric tensor (d_{ij}), the subscript i signifies the direction of the electric field displacement, whereas the subscript j represents the corresponding mechanical deformation.

Given that the current study is based on the stress form, the piezoelectric coefficients provided in Eq. 11 in strain form have been transformed into the stress form using the following relationship (Singh et al. 2020):

$$e_{ijk} = c_{jklm} d_{ilm}, \tag{12}$$

where e_{ijk} are the piezoelectric stress coefficients, c_{jklm} are the components of the elastic tensor, and d_{ilm} are the piezoelectric strain coefficients.

The orientation of the microtubules and their effect on the piezoelectric tensor have also been taken into account in this work. The transformation of the piezoelectric coefficients has been formulated elsewhere (Singh et al. 2020).

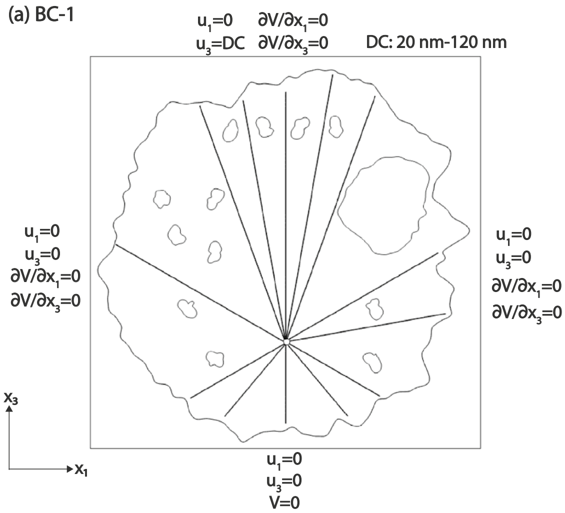
2.1 Boundary conditions

The details of the biological cell geometry and the governing equations to understand the piezoelectric behaviour of the biological cell are provided previously. Cells undergo compressive stresses from external contact and tensile forces exerted by neighbouring cells. The current work has focused on compressive displacements. To mimic a realistic and complex environment, a compressive displacement ranging from 20 to 120 nm is applied to the top, bottom, left, and right sides of the cell. Here, we discuss the impact of mechanical displacements on the biological cell, specifically examining three different boundary conditions to understand the cell response comprehensively, as shown in Table 2. The bottom surface of the two-dimensional biological cell model depicted in Fig. 2 has been subjected to fixed boundary conditions that are electrically grounded ($V = 0$). These restrictions have been adopted from previous studies (Garcia and Garcia 2018a, b; Li et al. 2018; Marcotti et al. 2019) and inspired by experimental investigations (Roggen et al. 1966; van Deurs 1973; Dallai et al. 1992; Mollaret and Justine 1997; Zhang and Assouline 2007; Lin et al. 2012; Hoyer-Fender 2013; Ishikawa 2017; Nakamura et al. 2021; Bezler et al. 2022). The direction in which the mechanical displacement is exerted on the piezoelectric material is as follows:

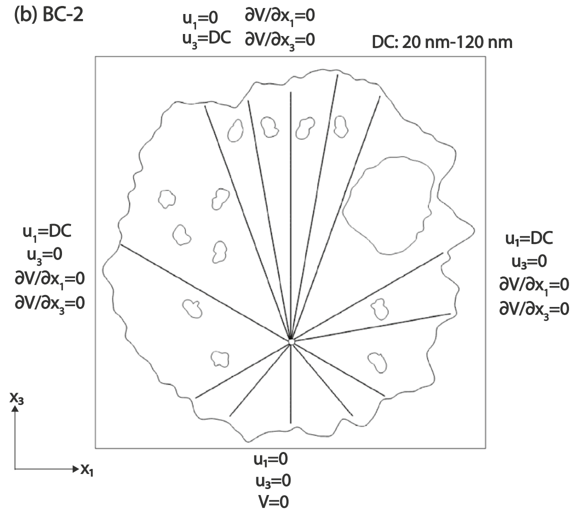
- Boundary condition-1 (BC-1): A mechanical displacement is exerted at the top of the boundary by altering the compressive displacement (DC) within the range of 20 nm to 120 nm, as depicted in Fig. 2(a).
- Boundary condition 2 (BC-2) involves the application of a mechanical displacement at the left, right, and top boundaries by altering the compressive displacement within the range of 20 nm to 120 nm, as depicted in Fig. 2(b).
- Boundary condition-3 (BC-3) deals with the imposition of a mechanical displacement at the left, right, top, and bottom boundaries of the system. This force is exerted by altering the compressive displacement values within the range of 20 nm to 120 nm, as depicted in Fig. 2(c).

These boundary conditions are very important to understand the cell mechanics in many applications, including bone repair (Wang 2007; Tichý et al. 2010; Nakamura et al. 2021; Zhang et al. 2021; Yang et al. 2022).

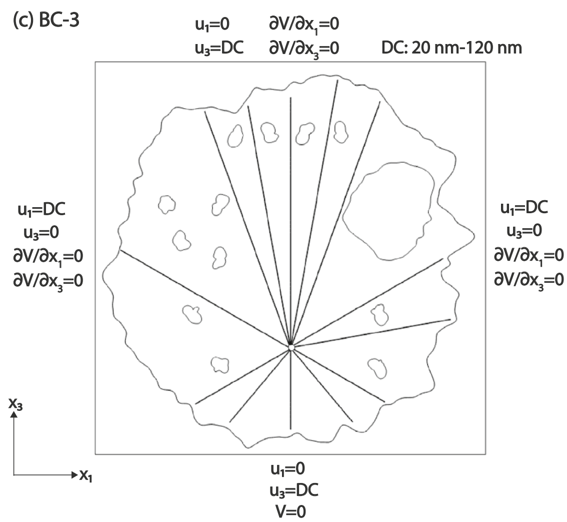
To investigate the mechanical response and piezoelectric behavior of the biological cell more precisely, effective coefficients have been determined by creating many load instances with varying boundary conditions. Each load instance should be structured in such a way that only one value in the strain/electric field vector is non-zero, with all other values being zero. To determine the effective coefficient, use an average non-zero value in the strain/electric field vector and the average values in the stress/electrical displacement vector (Berger et al. 2005; Dascalescu et al. 2014; Saputra et al. 2017; Hamdia 2024). The effective elastic coefficients and effective piezoelectric coefficients are computed in the principal plane directions (x_1-x_3) under the specified boundary conditions, i.e., the mechanical ($c_{11,eff}$, $c_{13,eff}$, and $c_{33,eff}$) and piezoelectric response ($e_{31,eff}$, and $e_{33,eff}$) of the biological cell in the transverse and longitudinal directions (Krishnaswamy et al. 2019a, c). The volume average of quantity A is denoted by $\langle A \rangle$ in the calculation below. It is determined as



(a) BC-1



(b) BC-2



(c) BC-3

Fig. 2 A finite element model of a single cell is created with specific boundary conditions (BC). Compressive displacement (DC) in terms of u_1 and u_3 is applied to the cell, ranging from 20 to 120 nm, in three different configurations: (**a**) on the top of the cell, (**b**) on the left and right sides of the cell, and (**c**) on the top, bottom, left, and right sides of the cell, while the bottom is electrically grounded in all the cases, i.e., the electric potential (V) is set to zero

Table 2 Boundary conditions considered in the current study

Boundary condition	Left	Right	Top	Bottom
BC-1	$u_1 = 0, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = DC, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = 0, V = 0$
BC-2	$u_1 = DC, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = DC, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = DC, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = 0, V = 0$
BC-3	$u_1 = DC, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = DC, u_3 = 0, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = DC, \frac{\partial V}{\partial x_1} = 0$	$u_1 = 0, u_3 = DC, V = 0$

$$\langle A \rangle = \frac{1}{(a_w a_w)} \int_{\Omega} A d\Omega, \tag{13}$$

where Ω represents the volume of the domain in which the integration is performed. A denotes the area of the biological cell, while a_w represents the width of the cell.

The effective elastic and piezoelectric coefficients are calculated by utilizing the volume averages for the boundary conditions specified in Sect. 2.1, which are as follows (Krishnaswamy et al. 2019a):

$$\begin{aligned} c_{11,eff} &= \frac{\langle \sigma_{11} \rangle}{\epsilon_{1\,1}}, \quad c_{13,eff} = \frac{\langle \sigma_{11} \rangle}{\epsilon_{3\,3}}, \quad c_{33,eff} = \frac{\langle \sigma_{33} \rangle}{\epsilon_{3\,3}}, \\ e_{31,eff} &= \frac{\langle D_3 \rangle}{\epsilon_{1\,1}}, \quad e_{33,eff} = \frac{\langle D_3 \rangle}{\epsilon_{3\,3}}, \end{aligned} \tag{14}$$

where $\langle D_3 \rangle$ is the volume average of the D_3 component of the electric flux density vector.

The finite element analysis has been performed to numerically implement the coupled electromechanical model described above using COMSOL multiphysics while applying boundary conditions to the biological cell.

3 Results and discussion

Cell mechanics refers to the division and merging of cell organelles, which are crucial for maintaining their structure, distribution, and size. In order to investigate the cell behaviour when it is subjected to mechanical displacement, three boundary conditions (BC-1 to BC-3) are employed for the compressive displacement ranging from 20 nm to 120 nm. The current study presents a comparative analysis of the piezoelectric effect in the cell, and analyzing the outcomes both with and without consideration of flexoelectric effects for the strain distribution, electric potential distribution, and determination of the effective elastic and piezoelectric coefficients.

Before reporting the new results from the current study, we verified its accuracy and reliability by conducting a comparative analysis with the existing literature (Singh et al. 2020), as shown in Fig. 3. It has been found that the current findings are consistent with the existing literature. The minor discrepancy in the results can be attributed to the increased number of mesh elements and fine mesh employed in the current work. A linear increase is observed in the maximal electric potential produced in the biological cell against the imposed compressive displacement. Moreover, the maximal electric potential in the cell increases proportionately with an increase in the number of microtubules from 10 to 18.

3.1 Impact of piezoelectric and flexoelectric effects on strain distribution in biological cells

The strain component (ϵ_{11}) distribution in the x_1-direction inside the biological cell as a result of the combined effects of piezoelectricity and flexoelectricity is shown in Fig. 4. The distribution has been derived for two distinct cell structures subjected to a compressive displacement of 20 nm, while accounting for three applied boundary conditions (BC-1 to BC-3). For BC-1, the magnitude of ϵ_{11} is concentrated in close proximity to the cell membrane, as well as the top and bottom walls of the cell, as depicted in Figs. 4(a), (b). Because the applied compressive mechanical displacement is from the top side only, i.e., the transverse direction. The strain distribution on left and right walls is

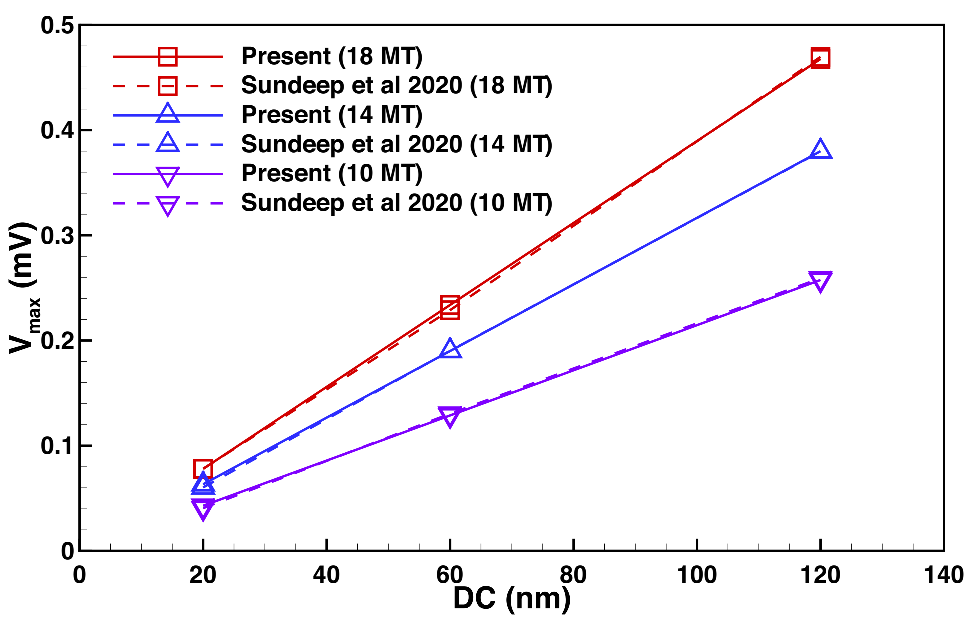


Fig. 3 A comparative analysis of the present results with existing literature (Singh et al. 2020)

negligible. Hence the resultant strain distribution concentrated in the close proximity of the top and bottom walls. For BC-2 and BC-3, the magnitude of ε_{11} is concentrated toward the corners of the bottom walls of the cell. However, the magnitude of the strain distribution in both CS-1 and CS-2 for BC-2 is approximately 10 times higher compared to the BC-3. Because the resultant strain for the BC-2 is concentrated close to the bottom walls and BC-3. Microtubules alone are the only components responsible for the piezoelectric and flexoelectric responses of the biological cell. As a result, we observe a significant variation in strain magnitude at the microtubule edges, at their junctions, and near their proximity. Except for these particular locations, the distribution of strain is rather consistent.

The strain distribution within the cell due to the combined of effects of the piezoelectricity and flexoelectricity is depicted in Fig. 5. It can be observed that there is a noticeable difference in the distribution of the strain component (ε_{33}) in the x_3-direction compared to the x_1-direction within the biological cell. This difference is attributed to the combined influence of piezoelectricity and flexoelectricity. The impact of flexoelectric stress components on the strain distribution is insignificant. Therefore, the strain distribution for the effect of piezoelectricity only and for the combined effect of piezoelectricity and flexoelectricity show a remarkable similarity and have nearly comparable magnitudes, as depicted in Figs. 4 to 5. Nevertheless, the strain distribution in the x_3-direction is slightly greater than in the x_1-direction, as the flexoelectric stress component in the x_3-direction is 1.80 times larger than its component in the x_1-direction, regardless of the boundary conditions (BC-1 to BC-3). Further, the flexoelectric stress component in cell structure-2 has exhibited a notable alteration in comparison to cell structure-1 as a result of the modifications in the shape, size, and orientation of the microorganelles, particularly the microtubules. Because the flexoelectric effect is basically shape dependent.

3.2 Electrical potential distribution

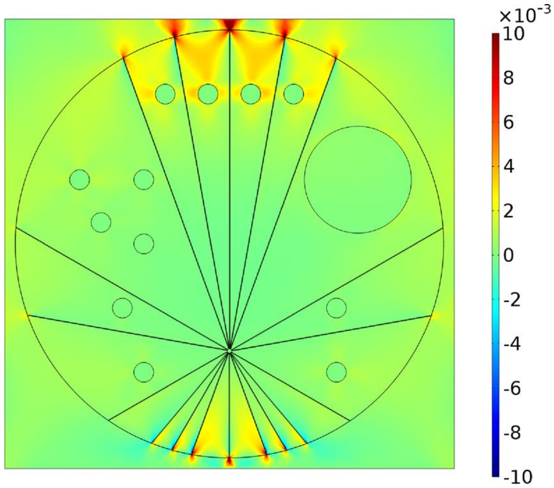
The electric potential distribution of a biological cell under compressive displacement has been determined by solving the governing equations described in Sect. 2. Accordingly, the electric potential distribution for each boundary condition applied to the cell considering only the piezoelectric effect and the combined piezoelectric-flexoelectric effects is provided below.

3.2.1 Consideration of piezoelectric effects alone

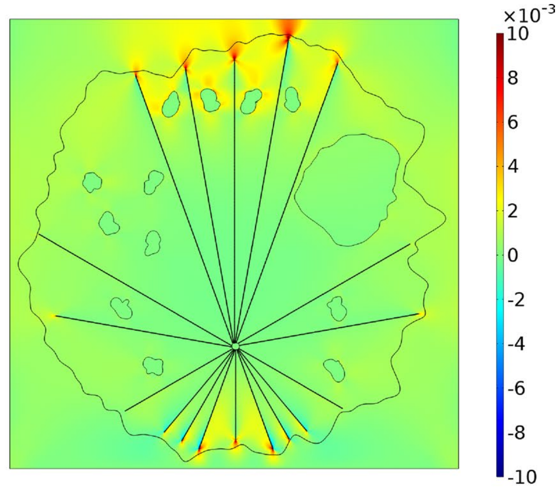
The electric potential (V) distribution within the biological cell, resulting from a 20 nm compressive mechanical displacement, accounting for the piezoelectric effect alone, is illustrated in Fig. 6. The distribution is studied under three applied boundary conditions, marked as BC-1 to BC-3. The arrows indicate the direction of the electric field lines within the cell. In all the cases, BC-1 and BC-3, it is evident from Figs. 6(a)-(d) that the electric potential generated is higher for CS-1 in comparison to CS-2. Because the strain distribution is no more streamlined along the boundary of the microtubules due to the localization of the strain near to the curved edges. As a result, the intensity of the strain distribution in the vicinity of the centrosome is reduced. Hence, the overall electric potential being generated in the cell structure-2 is reduced compared to the cell structure-1. Consequently, a weak piezoelectric effect leads to the generation of a lower electric potential. The electric potential being generated has been reduced by a factor of 10 when BC-3 is applied to the biological cell compared to BC-1 and BC-2 for both CS-1 and CS-2 (see Figs. 6(e)-(f)). For regular shapes, the distribution of the electric field lines is symmetrical. Because the microtubules are symmetrically distributed along the centerline of the x_3-axis. On the contrary, when dealing with an irregular shape geometry, the electric field lines exhibit greater intensity at the right corner compared to the left corner. This results in the production of a greater amount of electric potential. Furthermore, the electric potential gradient is higher in regular shape geometry (CS-1) because it contributes to more mechanical strain production. Therefore, CS-1 overpredicts the response to external mechanical forces.

3.2.2 Consideration of flexoelectric effects in addition to piezoelectric effects

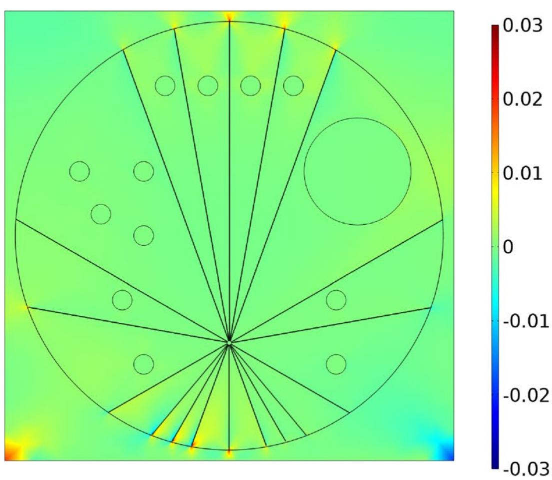
There is a significant enhancement of the electric potential distribution within the cell when considering the application of non-local flexoelectric effects in addition to the linear piezoelectric effects. Figs. 7(a), 7(c) and 7(e) depict the electric potential distribution in CS-1, whereas Figs. 7(b), 7(d) and 7(f) for the CS-2. The contour plots displayed in Figs. 7(a)-(d) demonstrate that, under all three boundary conditions (BC-1 to BC-3), the electric potential generated in CS-1 is greater than that of CS-2. The distribution of the electric field lines is symmetrical under BC-1 for the CS-1 and the strain gradients are stronger for the regular shapes of organelles in the cell (CS-1). When dealing with CS-2, the electric field lines also exhibit a symmetrical about the centerline x_3-axis but due to the irregular shapes of the organelles, such as the cytoplasm, cell membrane and microtubules. Consequently, this leads to the generation of a higher magnitude of electrical potential. Moreover, the electric potential gradient is greater in CS-1 due to its contribution to increased mechanical strain generation. The electric potential generated in BC-2 is observed to increase due to



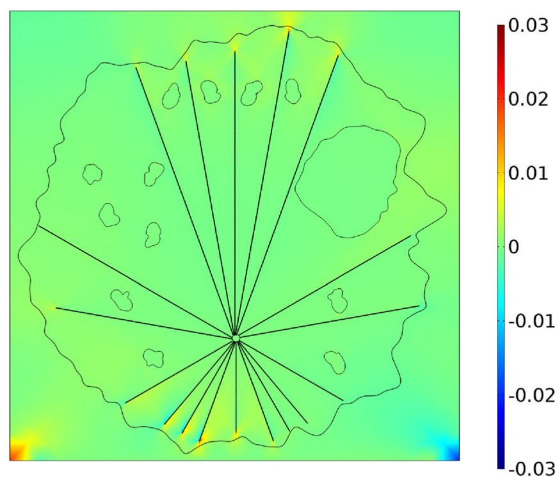
(a) CS-1, BC-1



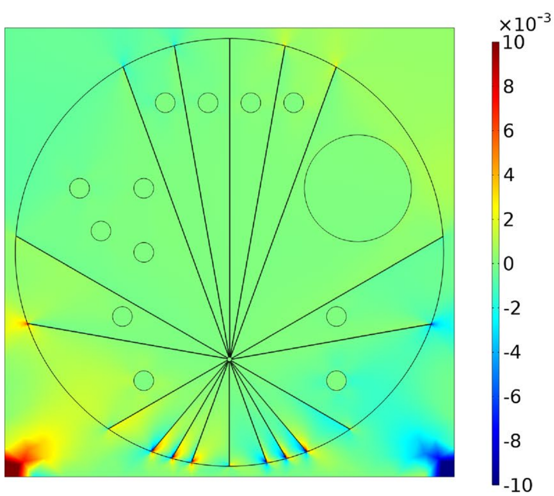
(b) CS-2, BC-1



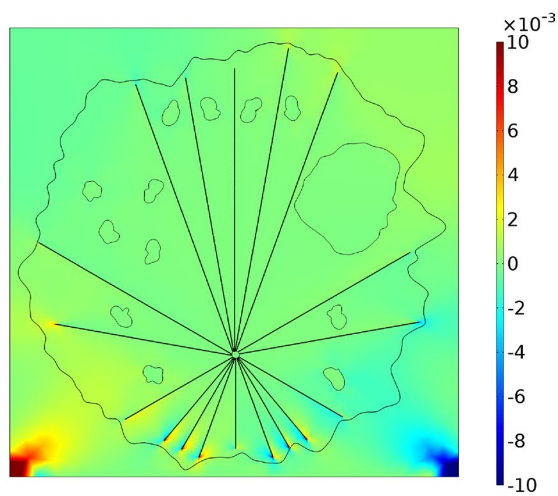
(c) CS-1, BC-2



(d) CS-2, BC-2



(e) CS-1, BC-3



(f) CS-2, BC-3

◂**Fig. 4** The distribution of the strain component in the x_1-direction (ε_{11}) for two cell structures (CS-1 and CS-2), namely cell structure-1 (a, c, e) and cell structure-2 (b, d, f), under three different boundary conditions considering the piezoelectric effect alone and the combined effects of the piezoelectricity and flexoelectricity for the compressive displacement of 20 nm

the absence of strain gradients in the longitudinal directions. The transverse direction (x_3) exhibits a strain gradient that is responsible for the increased electric potential compared to BC-1 and BC-3. Overall, the combined effect of piezoelectricity and flexoelectricity induces a higher electric potential when considering only the piezoelectric effect for both the cell structures.

3.2.3 Consideration of the number of microtubules and the shape of the organelles

It is observed that the maximum electrical potential obtained for each boundary condition and cell structure is also different, as depicted in Figs. 6 and 7. The maximum electric potential value is located at the microtubule edges within the biological cell, consistent with the findings from earlier studies (Havelka et al. 2011). This is attributed to the shape and orientation of the microtubules within the biological cell that have a significant impact on the distribution of the electric potential (Denning et al. 2017). The absence of longitudinal and transverse strains in BC-3 can be attributed to the uniform displacement boundary conditions in the principal planes, i.e., x_1 and x_3. Nevertheless, only a small amount of transverse strain is generated near the corners of the bottom side due to the applied mechanical displacement. This can be attributed to the negligible piezoelectric effect, resulting in a low generation of electric potential in this case. The increase in the number of microtubules leads to the higher electric potential generation in the cell. These findings on the electric potential demonstrate the importance of considering both piezoelectric on the electro-elastic properties of living cells, particularly when considering cellular length scales.

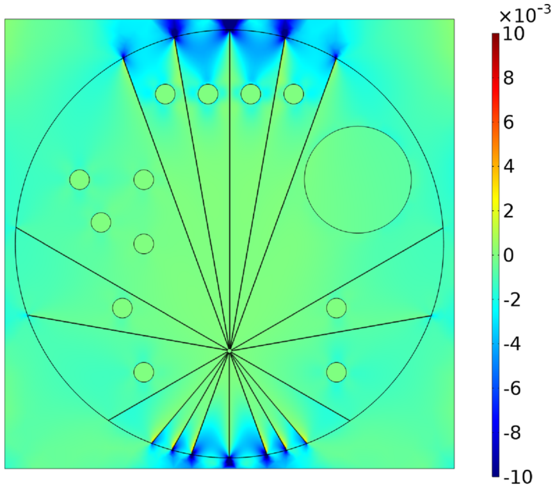
The growth, maintenance, and regulation of cells and tissues depend on the strength of electric fields. Moreover, studying electric potential distribution within the cell is essential for tissue regeneration, embryonic development, and wound healing (Titushkin and Cho 2009; Schofield et al. 2020). For instance, electric fields are used in tissue engineering for characterizing artificial tissues, forming tissue-like materials, and micro-manipulating cells. Electrical stimulation has emerged as a novel and cutting-edge method in the field of regenerative medicine (Markx 2008; Chen et al. 2019).

3.2.4 Maximum induced electric potential considering the piezoelectric effect and the combined piezoelectric-flexoelectric effects

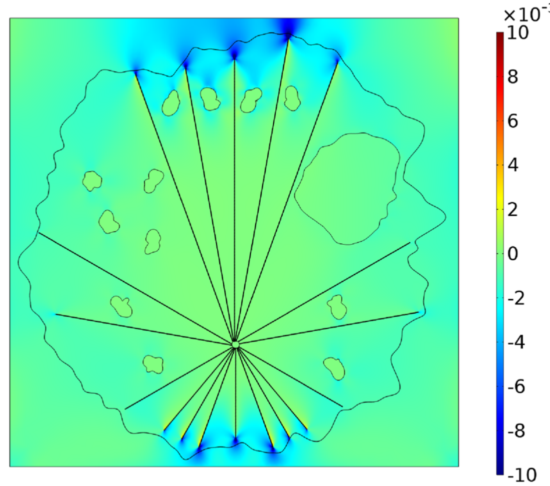
A parametric analysis has been carried out to quantify the maximum electric potential (V_{max}) obtained when a compressive displacement of 20 nm is applied for BC-1 to BC-3. The number of microtubules in the present two-dimensional biological cell considered are 18, as shown in Fig. 8. When considering purely the piezoelectric effect, CS-1 generates a V_{max} larger than CS-2 for BC-1, as illustrated in Fig. 8(a). On the other hand, V_{max} shows a significant rise in the case of CS-1 when both the piezoelectric and flexoelectric effects are taken into account together for BC-1, as shown in Fig. 8(b) and a similar pattern is observed for BC-2 and BC-3. An increase in the number of microtubules in the cellular structure results in an increase of V_{max}. This increase is evident from the compressive displacement range of 20 nm to 120 nm observed in BC-1 to BC-3, as shown in Figs. 8(a), (f). This occurs due to the symmetry of electric field lines along the x_3 centerline axis for CS-1. Whereas for CS-2, the orientation and shape of the microtubules are different, the maximum value of the electric potential being generated is also less compared to the CS-1. Recall that, in the present study, the piezoelectric and flexoelectric coefficients are considered only for the microtubules.

Figure 9 illustrates the variation in V_{max} distribution resulting from the piezoelectric-flexoelectric effects within the biological cell, while maintaining boundary conditions and cell structures. This variation is observed as a function of the compressive displacement and the number of microtubules. In CS-1 for BC-1, the induced V_{max} is significantly higher for the piezoelectric-flexoelectric effects compared to the piezoelectric effect alone. It can be observed from Figs. 9(a), (f) that an increase in the number of microtubules leads to an increase in the generated V_{max} for all the boundary conditions. This is attributed due to the orientation of the microtubules in the cells. Experimentally, it has been found that the orientation and alignment of microtubules significantly impact the shape and functionality of keratinocyte cells (Kobori et al. 2024).

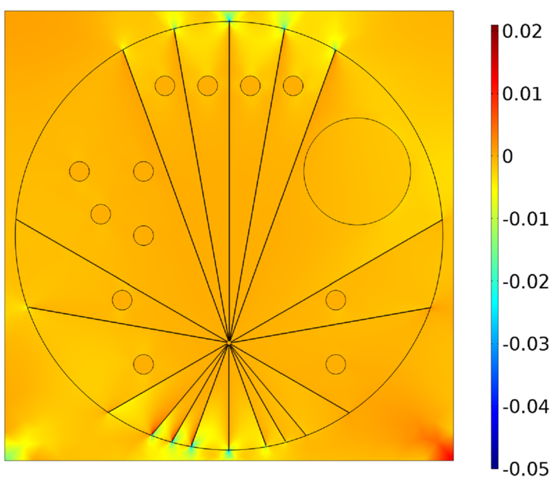
Figures 8 and 9 also clearly demonstrate that the magnitude of V_{max} follows a trend of increasing from BC-1 to BC-2 and thereafter decreasing at BC-3. For instance, when the piezoelectricity alone is considered, V_{max} for CS-1 is found to be 1.18×10^{-6} V (BC-1), 1.74×10^{-6} V (BC-2), and 4.92×10^{-7} V (BC-3). Experimental findings confirm that the maximum induced electric potential in the bone by the piezoelectric effect is on the order of microvolts (μV) (Nakamura et al. 2021). Further, a similar pattern has been noticed in the case of CS-2. This occurs because, in BC-1, the electric potential is solely generated by the transverse strain from the top side, without any contribution from the



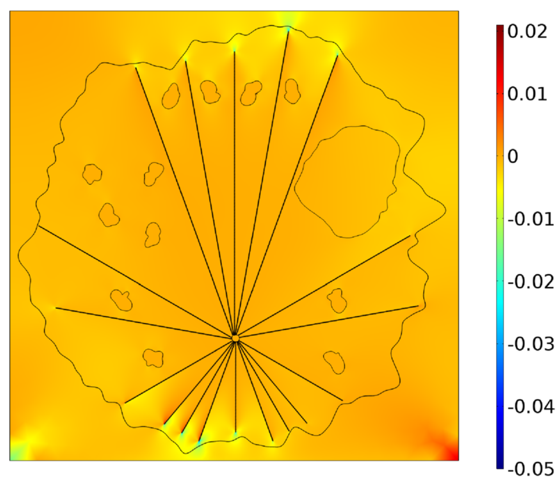
(a) CS-1, BC-1



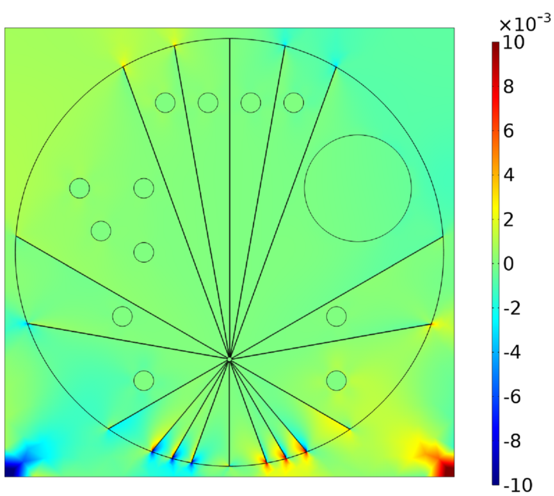
(b) CS-2, BC-1



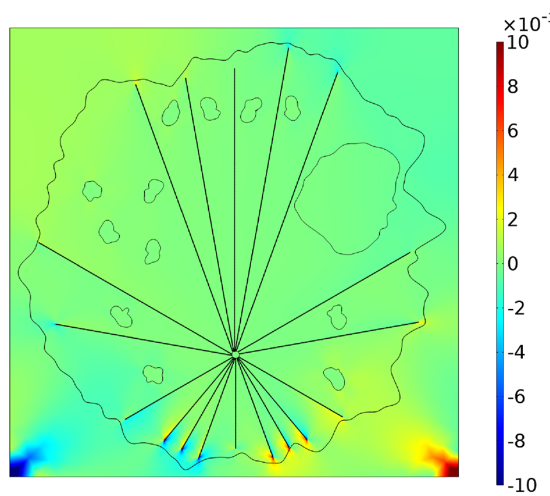
(c) CS-1, BC-2



(d) CS-2, BC-2



(e) CS-1, BC-3



(f) CS-2, BC-3

◂**Fig. 5** The distribution of the strain component in the x_3-direction (ε_{33}) for the piezoelectric effect alone and the combined effects of piezoelectricity and flexoelectricity for cell structure-1 (a, c, e) and cell structure-2 (b, d, f) under three different boundary conditions for the compressive displacement of 20 nm

longitudinal strain. In BC-2, the applied compressive displacement on both the left and right sides is balanced due to the absence of a strain gradient. The gradient of BC-3 is symmetrical from all directions, resulting in a minimal net value of the electric potential.

3.2.5 The maximum electric potential ratio

Flexoelectricity exerts a substantial influence on the biological cell. It is additionally characterized by the ratio of the maximum electric potential resulting from the combined effects of piezoelectricity and flexoelectricity to the electric potential arising from piezoelectricity alone ($V_{\mathrm{R,max}} = V_{\mathrm{max,P+F}}/V_{\mathrm{max,P}}$). This ratio has been examined in connection with the total number of microtubules under the three specified boundary conditions, as illustrated in Fig. 10. It is evident that the highest value of $V_{\mathrm{R,max}}$ occurs when there are 6 microtubules in the case of CS-2, under the three boundary conditions BC-1 to BC-3. It has been found that when BC-1 is applied, $V_{\mathrm{R,max}}$ for CS-1 takes precedence over CS-2. Whereas $V_{\mathrm{R,max}}$ has shown an uneven trend for CS-1 on imposing BC-2 and BC-3. The primary factor influencing the behaviour is the angular orientation and shape of the microtubules (Böhm et al. 2005; Nakamura et al. 2021).

This knowledge is very useful to our better understanding of the induced electric potential resulting from piezoelectricity in the healing mechanism for bone fractures and other medical therapies. For example, the primary building blocks of bone, collagen and hydroxyapatite, are oriented strongly in the direction of the bone axis. The direction in which ultrasonic radiation or a load is applied greatly influences the induced electric potential in the bone resulting from piezoelectricity. The piezoelectric properties of bone have a role in the healing process of the Low-intensity pulsed ultrasound (LIPUS) therapy (Fukada and Yasuda 1957; Antebi et al. 2012; Nakamura et al. 2021). The two-dimensional biological cell is significantly influenced by the quantity and loading conditions of microtubules. The current study predicts a significant increase in electromechanical effects, potentially useful in bio-compatible nano-biosensors, drug delivery, noninvasive diagnostic methods, and treatment strategies (Deng et al. 2014; Peppas and Clegg 2016; Ashammakhi et al. 2018; Kamel 2022).

The stimulation of cells depends on the generation of electric potential resulting from the mechanical load being applied. For instance, the glial cells undergo changes in structure when stimulated electrically and align themselves perpendicular or parallel to the electric field (Tsui et al. 2022). Hence, the biological cell stimulation depends on the intensity of the electric potential, which is dependent on the boundary condition being imposed. The biological cell having a structure with regular-shaped organelles exhibited over cellular stimulation than the arbitrary shapes of the organelles. The previous study utilized a cell structure (Singh et al. 2020) model that exhibited regular shapes of the organelles. Therefore, in the present study, we carry out a more realistic analysis where the shapes of the organelles have been chosen to be arbitrary, which are non-spherical in nature. Moreover, the cell stimulation is controlled by varying the mechanical displacement as described previously (Sect. 2.1).

3.3 Determination of the effective elastic coefficients

The computational model employed in this work represents the biological cell, which consists of several organelles. Since each organelle has unique properties, it is crucial to understand the overall behaviour of the biological cell. This can be achieved by determining the effective elastic and effective piezoelectric coefficients of the biological cell. Effective elastic coefficients are computed using Eqs. 13 and 14 to elucidate the mechanical response of the cell, which show that the principal plane stresses are the only factors influencing these coefficients. The compressive displacement has an impact on the effective elastic coefficients, $c_{11,eff}$ and $c_{33,eff}$ in the principal planes (x_1 and x_3). We also have another elastic coefficient, $c_{13,eff}$, defined as the ratio of the average principal stress in the x_3-direction to the principal strain in the x_1-direction. The following sub-section discusses the numerical studies conducted to investigate the variations in the effective elastic coefficients influenced by the structure of biological cell and physical properties, considering only the piezoelectric and combined piezoelectric-flexoelectric effects.

3.3.1 Consideration of piezoelectric effects alone and combination of piezoelectric-flexoelectric effects

The effective elastic coefficient ($c_{11,eff}$) is determined when the mechanical displacement is applied in the x_1 direction, as shown in Fig. 11(a). It has been found that under BC-1, the value of $c_{11,eff}$ is approximately 100 times greater for CS-1 than for CS-2, considering solely the piezoelectric effect. Moreover, it is important to mention that the value of $c_{11,eff}$ remains similar for both the piezoelectric and flexoelectric effects. As a result, the flexoelectric effect is negligible. Differences in the shapes of the organelles do not affect the effective elastic coefficient. Nevertheless, the value of $c_{11,eff}$ exhibits a declining pattern as the number of microtubules

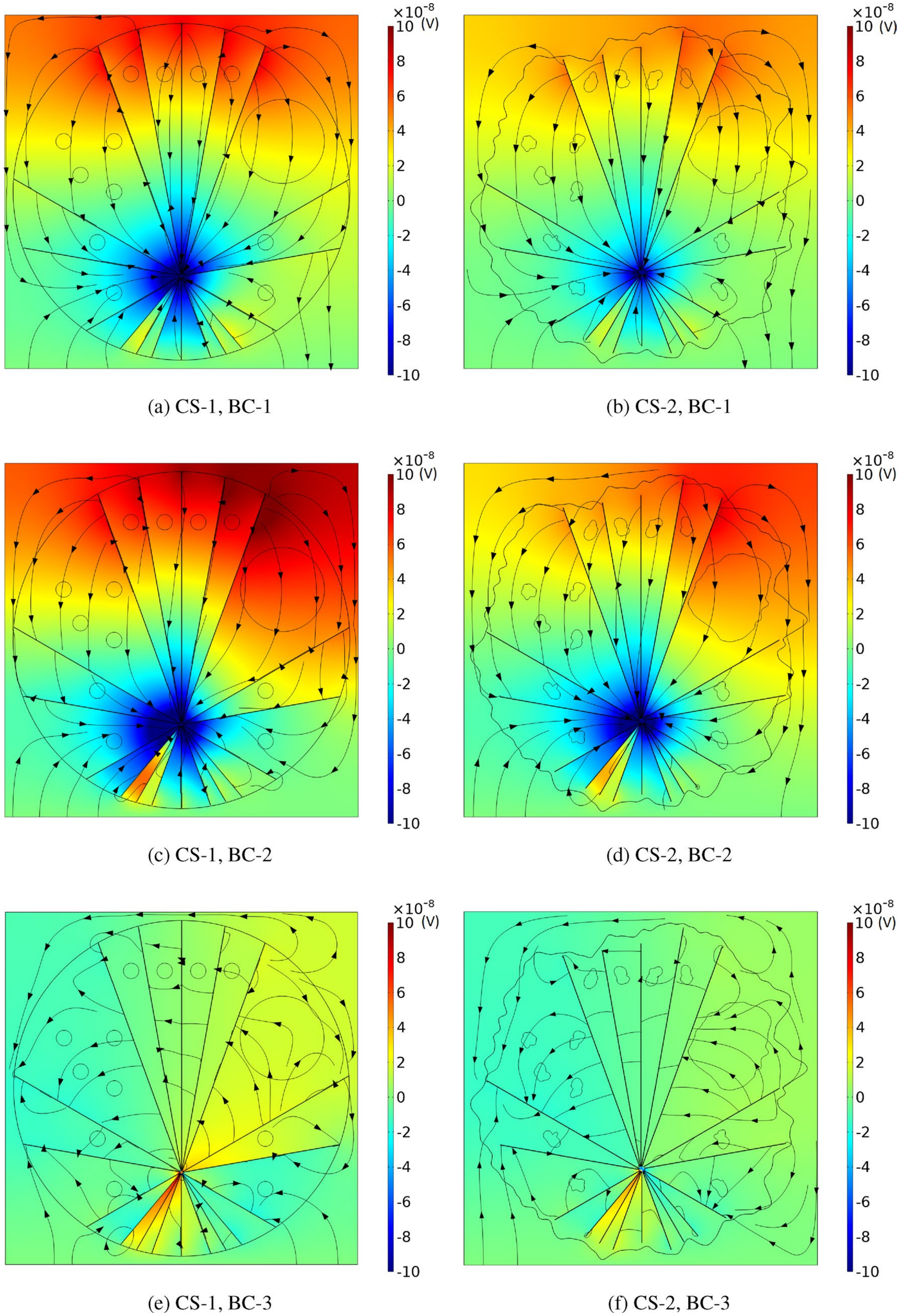


Fig. 6 The electric potential distribution in the biological cell is being analyzed under the influence of a 20nm compressive displacement, taking into consideration only the piezoelectric effect. The analysis has been conducted for two cell structures: CS-1 (a, c, e) and CS-2 (b, d, f)

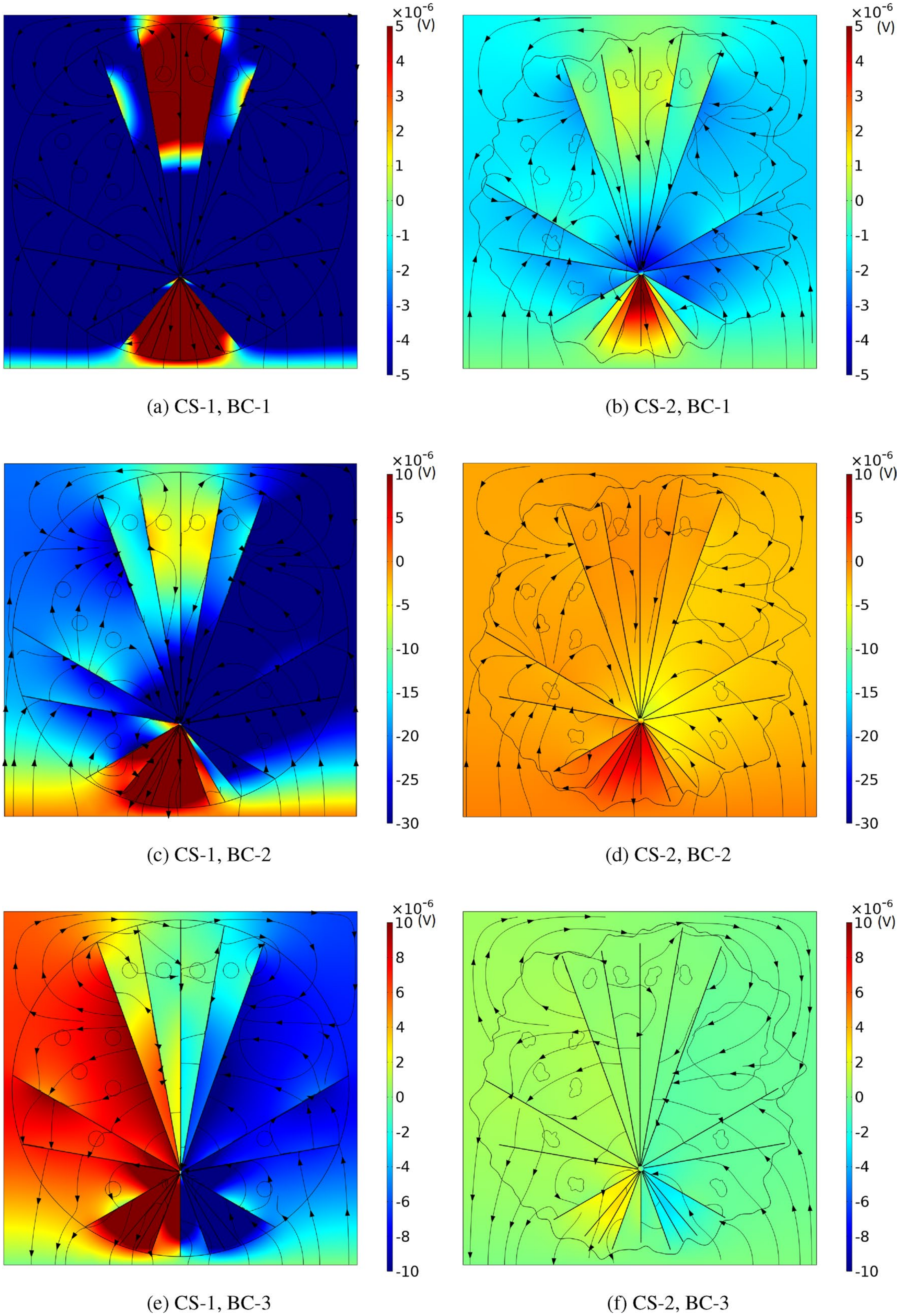
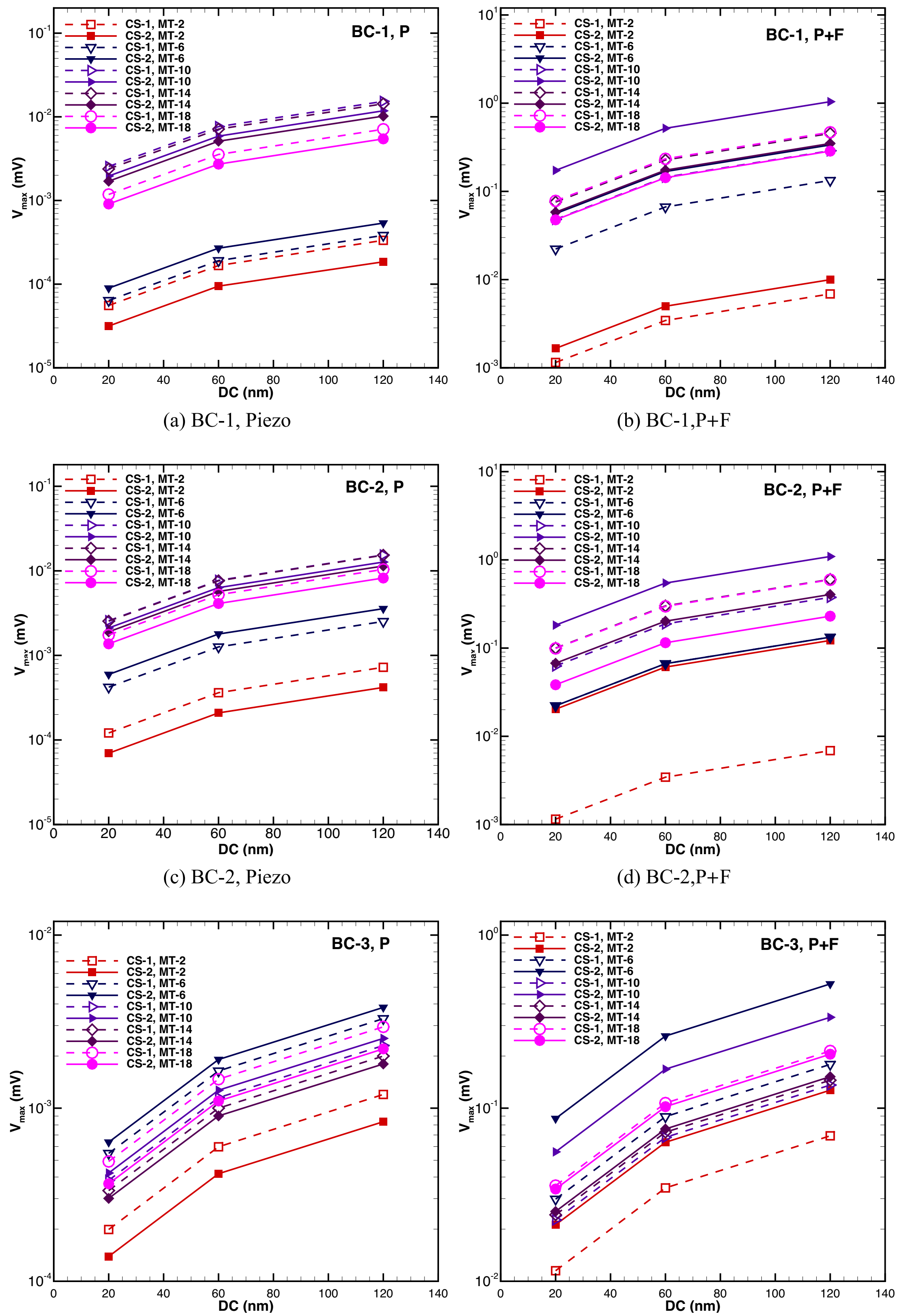


Fig. 7 The electric potential distribution in the biological cell is analyzed for a compressive displacement of 20 nm, taking into account both the piezoelectric and flexoelectric effects for two cell structures: CS-1 (a, c, e) and CS-2 (b, d, f)

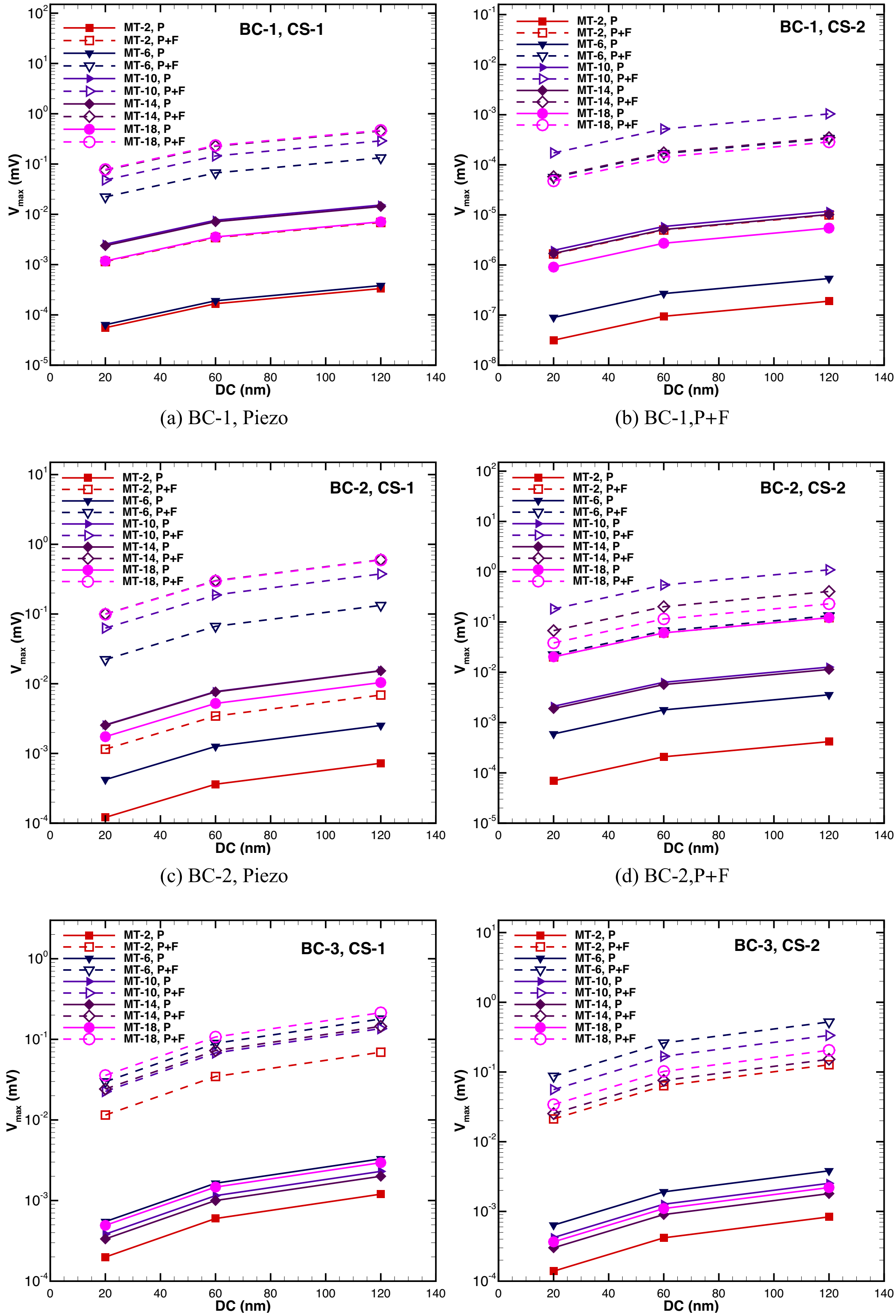


(a) BC-1, Piezo (b) BC-1,P+F

(c) BC-2, Piezo (d) BC-2,P+F

(e) BC-3, Piezo (f) BC-3,P+F

Fig. 8 Maximum electric potential (V_{max}) as a function of the compressive displacement and number of microtubules considering only for the piezoelectric effect (a,c,e) and for the combined effect of piezoelectricity-flexoelectricity (b, d, f) under the specific boundary conditions



(a) BC-1, Piezo

(b) BC-1,P+F

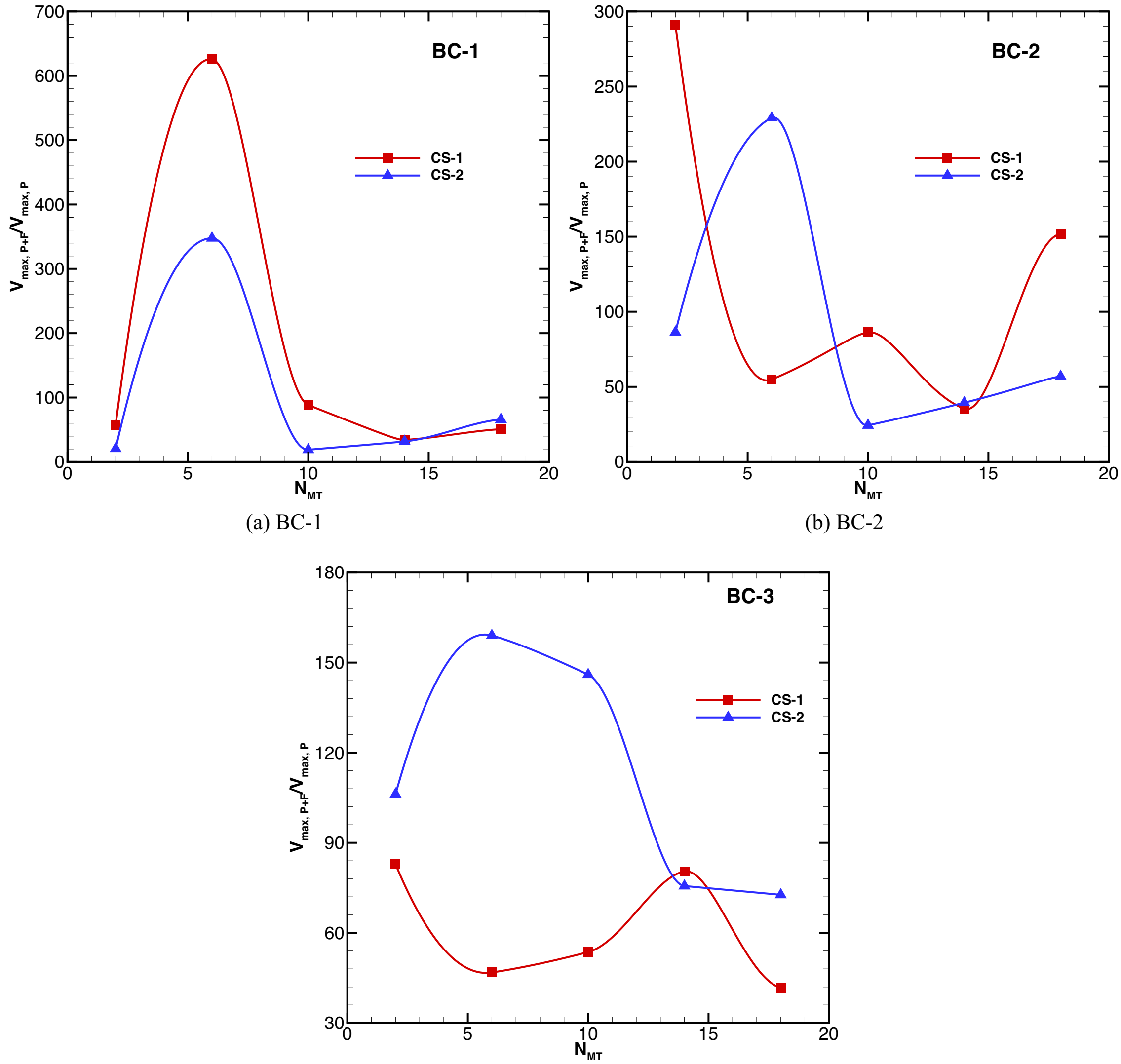
(c) BC-2, Piezo

(d) BC-2,P+F

(e) BC-3, Piezo

(f) BC-3,P+F

Fig. 9 Maximum electric potential distribution (V_{max}) and the compressive displacement and number of microtubules in the case of the piezoelectric (a,c,e) and flexoelectric (b, d, f) effects, considering the three boundary conditions



(a) BC-1 (b) BC-2

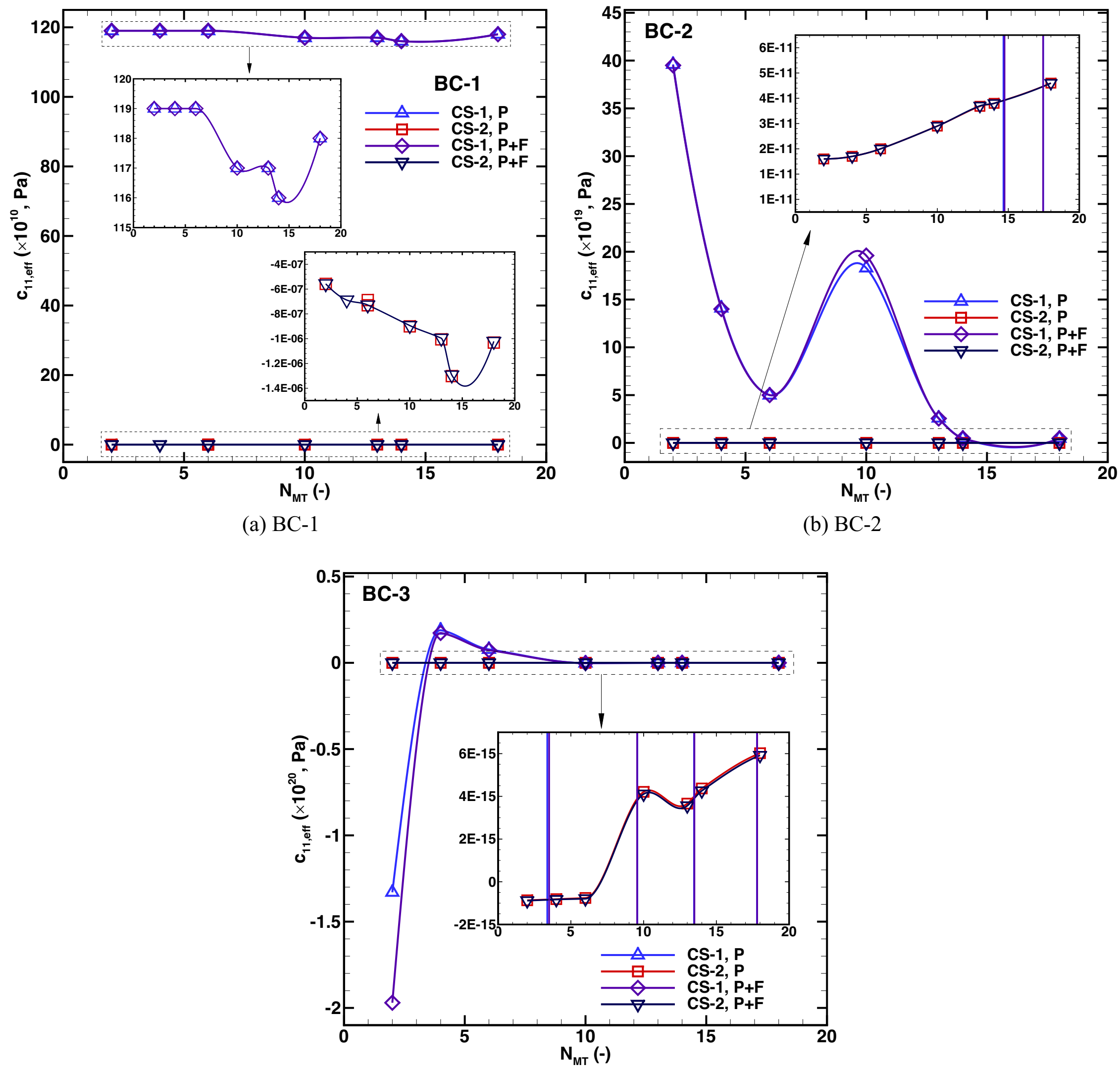
(c) BC-3

Fig. 10 Ratio of the maximum electric potential for the piezoelectric and flexoelectric effects to the piezoelectric effects, denoted as $V_{\mathrm{R,max}} = V_{\mathrm{max,P+F}}/V_{\mathrm{max,P}}$, as a function of the number of microtubules under three boundary conditions

(N_{MT}) increases. This phenomenon can be attributed to the consideration of the piezoelectric properties of the microtubules and their alignment within the cell.

For BC-2, there is no fluctuation in the value of $c_{11,eff}$ for CS-2 when only the piezoelectric effect has been taken into account. However, the value of $c_{11,eff}$ is lower for CS-1 compared to CS-2 and displays variation for CS-1, as depicted in Fig. 11(b). The variations in the shapes of the organelles result in a decrease in the individual contribution for c_{11} for CS-2 as determined against CS-1. For this particular boundary condition, the compressive displacement is exerted on three sides of the cell, while the bottom side is electrically grounded. Therefore, there is no variation in stress along the x_3-direction, and it is only present in the x_1-direction. $c_{11,eff}$ exhibits a declining trend in the context of the combined piezoelectric and flexoelectric phenomena as the number of microtubules is augmented from 2 to 6. The cell effective behavior when the mechanical load is imposed in the x_1-axis is dependent on the number of microtubules. However, it experiences a rapid surge at 10 microtubules, followed by a subsequent decrease see Fig. 11(b).

Upon analyzing Fig. 11(c), it becomes apparent that when the compressive displacement is applied to all sides of the cell (BC-3), the effective coefficient of $c_{11,eff}$ displays lower values for CS-1 initially, but eventually reaches the same value as CS-2 when just the piezoelectric effect is taken into account. The effective elastic coefficient $c_{11,eff}$ is larger for CS-1 compared to CS-2, particularly for up to 4 microtubules, considering the combination of the piezoelectric and flexoelectric effects, and $c_{11,eff}$ shows a declining pattern in



(a) BC-1

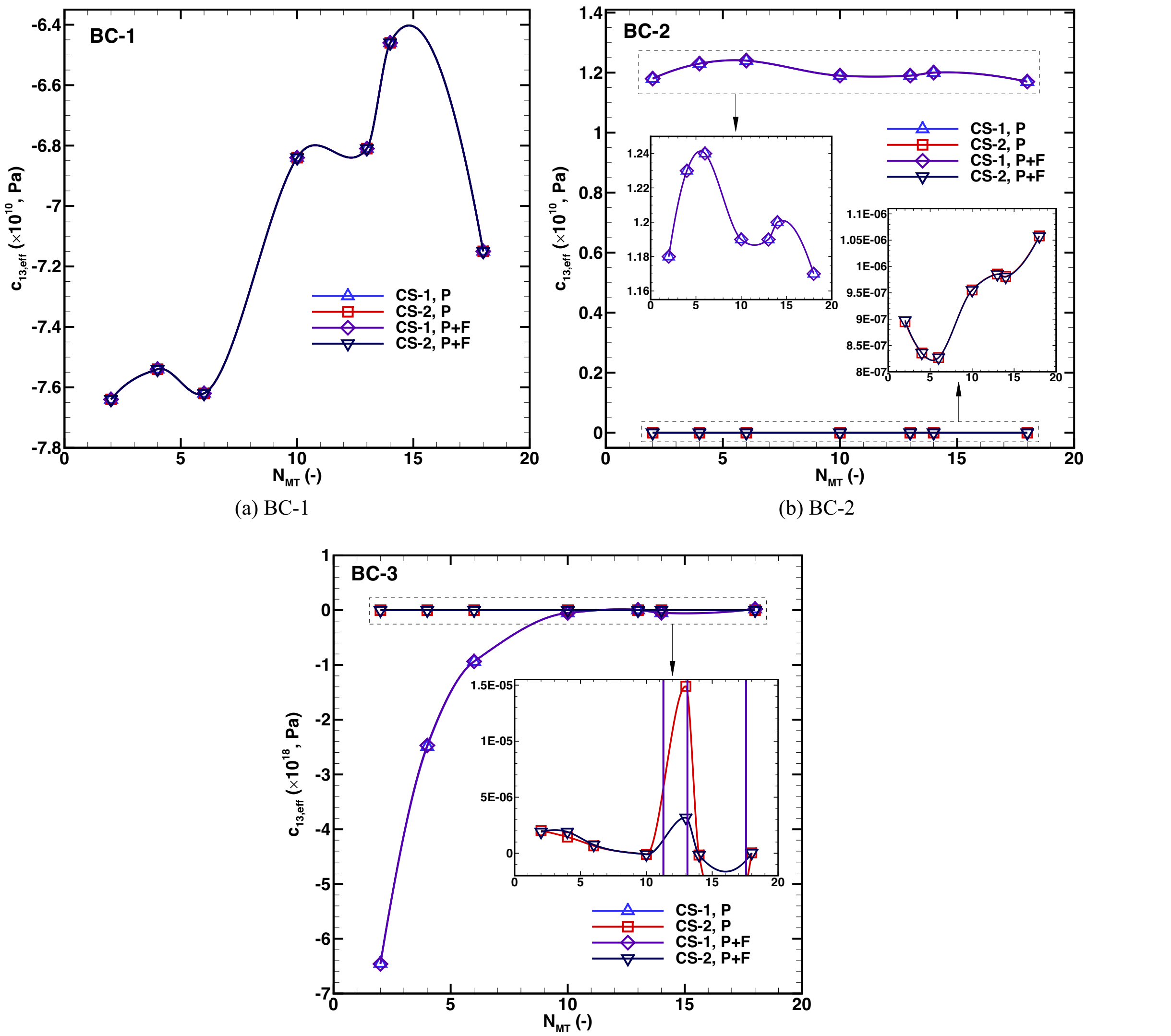
(b) BC-2

(c) BC-3

Fig. 11 The effective elastic coefficient ($c_{11,eff}$) of cell structure-1 (CS-1) and cell structure-2 (CS-2) as a function of the number of microtubules for the piezoelectric (P) and flexoelectric (P+F) effects are plotted against the compressive displacement at 20 nm for (a) BC-1, (b) BC-2, and (c) BC-3

CS-1. This drop is noticed when the number of microtubules is increased from 4 to 6. Moreover, $c_{11,eff}$ converges to a comparable value as that of CS-2 when the number of microtubules surpasses 10. This phenomenon occurs because the strain generated can be seen only in the x_1-direction, as a result of the mechanical displacement. Furthermore, the impact of the flexoelectric effect is insignificant, even for BC-3.

Figure 12 depicts the changes in $c_{13,eff}$ across the biological cell, as computed by taking into account the piezoelectric effects. For BC-1, it is apparent from Fig. 12a that $c_{13,eff}$ exhibits a consistent pattern for both CS-1 and CS-2 when only the piezoelectric effect is considered. However, it is evident from Fig. 12(b) that $c_{13,eff}$ value remains unchanged for both the piezoelectric and flexoelectric effects. $c_{13,eff}$ is shown to decrease initially as the number of microtubules is increased from 2 to 6. However, the effect of the shape change and flexoelectricity is insignificant. This is because the average stress values in the x_3-axis and strain in the x_1 -direction are the same for both cell structures. In the case of BC-2, $c_{13,eff}$ is higher for CS-2 in comparison to CS-1 because the principal stresses and strains for these cell structures are different, i.e., the shape changes of the organelles. Moreover, the value of $c_{13,eff}$ remains unchanged for both the piezoelectric and flexoelectric effects, as depicted in Fig. 12(b). The impact of the flexoelectricity is negligible. It can be observed that the value of $c_{13,eff}$ shows no variation while the number of microtubules is increased from 2 to 6.



(a) BC-1

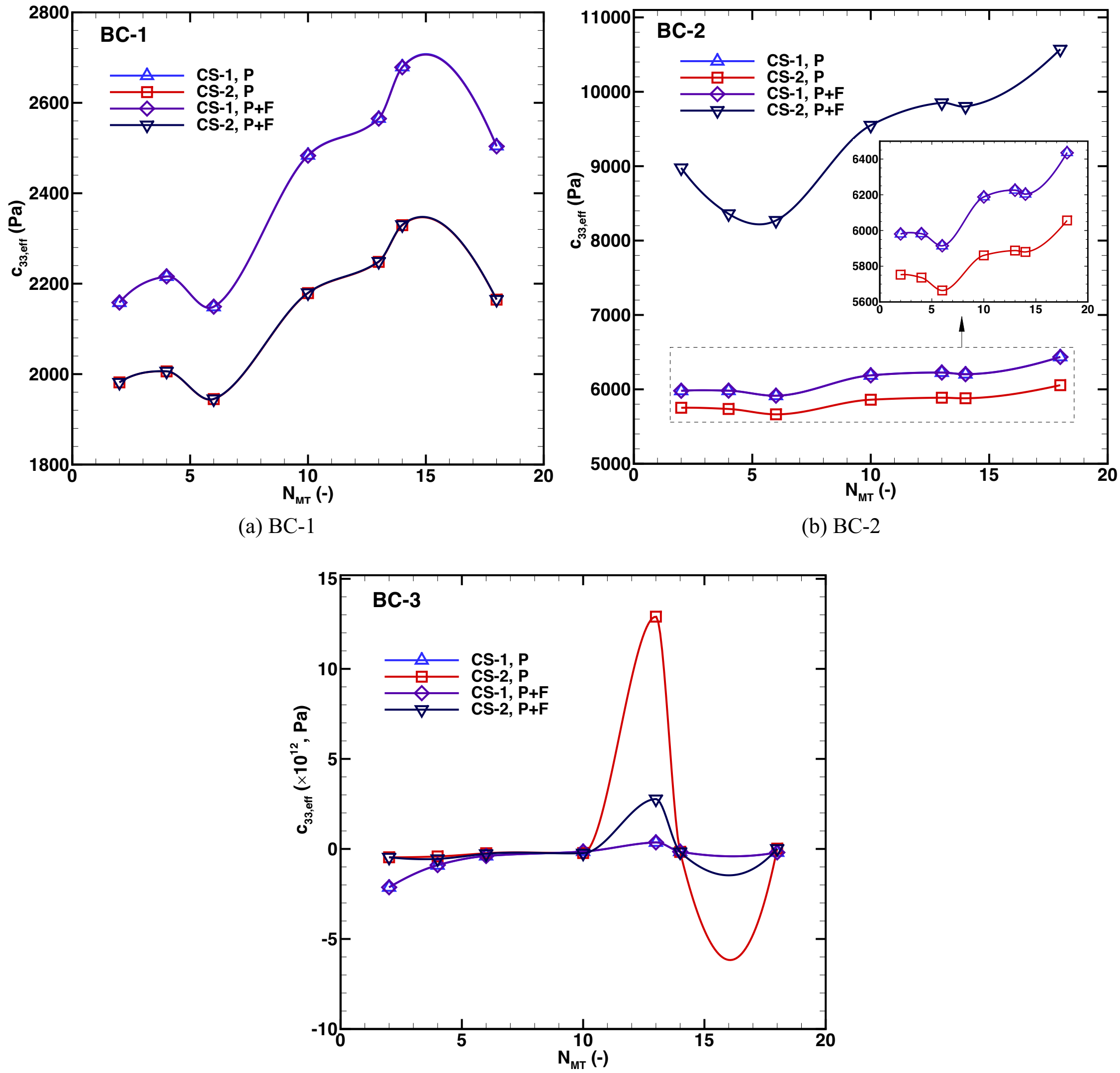
(b) BC-2

(c) BC-3

Fig. 12 The effective elastic coefficient ($c_{13,eff}$) of cell structure-1 (CS-1) and cell structure-2 (CS-2) as a function of the number of microtubules considering only the piezoelectric effect (P) and for combined piezoelectric-flexoelectric (P+F) effects plotted against the compressive displacement at 20 nm for (a) BC-1, (b) BC-2, and (c) BC-3

However, there is a dramatic decrease in $c_{13,eff}$ thereafter, followed by a period of no apparent variation. Therefore, the variation in the value of $c_{13,eff}$ for an individual cell structure is mainly due to the number of microtubules present within the biological cell. In the case of BC-3, $c_{13,eff}$ value is lower for CS-2 in comparison to CS-1, specifically for up to 10 microtubules, when considering the piezoelectric and flexoelectric effects. Subsequently, the value of $c_{13,eff}$ tends to converge with that of CS-1 when considering 10 microtubules and beyond, as depicted in Fig. 12(c). Because the overall response of the cell for $c_{13,eff}$ is increasing primarily due to the quantity of the microtubules and its shape.

Figure 13(a) shows that the effective value of c_{33} (the mechanical response of the biological cell in x_3-direction) is greater for CS-1 than for CS-2 under BC-1. Both cell structures exhibit an increasing trend in $c_{33,eff}$ with an increase in the number of microtubules. $c_{33,eff}$ value remains unchanged for both the piezoelectric and flexoelectric phenomena. The value of $c_{33,eff}$ exhibits an upward trend when the number of microtubules increases from 2 to 4. However, it experiences an abrupt decline when the number of microtubules reaches 6. The values of $c_{33,eff}$ are shown to increase for both the piezoelectric and flexoelectric effects, as well as for CS-1 and CS-2, extending beyond microtubules 6. The stress and strain are present only in the x_1- and x_3-directions. It is clearly evident that the shape of the organelles plays an important role. The regular shapes of the organelles (CS-1) over predicts the



(a) BC-1

(b) BC-2

(c) BC-3

Fig. 13 The effective elastic coefficient ($c_{33,eff}$) of cell structure-1 (CS-1) and cell structure-2 (CS-2) as a function of the number of microtubules considering only the piezoelectric effect (P) and for combined piezoelectric-flexoelectric (P+F) effects are plotted against the compressive displacement at 20 nm for (a) BC-1, (b) BC-2, and (c) BC-3

value of $c_{33,eff}$ compared to the irregular shapes of the organelles (CS-2) As seen in Fig. 13(b), both cell structures exhibit a consistent pattern with an increase in microtubule number for BC-2 when considering the piezoelectric effect alone, and $c_{33,eff}$ is slightly larger for CS-1 than CS-2 due to the symmetrical geometry in x_3-direction. It is apparent that $c_{33,eff}$ value remains unchanged for both the piezoelectric and flexoelectric phenomena. $c_{33,eff}$ is declining as the number of microtubules increases from 2 to 6 due to the orientation of the microtubules. However, it experiences a dramatic decline at 10 microtubules. The values of $c_{33,eff}$ are observed to increase for both the piezoelectric and flexoelectric effects considering the cell structures, CS-1 and CS-2, extending the microtubules beyond 10. In the case of BC-3, it is evident that $c_{33,eff}$ is comparatively lower for CS-1 than for CS-1, specifically within the range of 6 microtubules, with regard to the piezoelectric and flexoelectric phenomena. The value of $c_{33,eff}$ is shown to converge with that of CS-1 when considering 6 microtubules, as depicted in Fig. 13(c). In addition, when the number of microtubules are 10, it is seen that the effective value of $c_{33,eff}$ for CS-1 starts rapidly increasing. The cell behavior in terms of $c_{33,eff}$ is approaching a constant value shows that the effect of piezoelectricity, shape variation of the organelles is insignificant at higher microtubules due to their orientation within in the cell.

3.3.2 Impact of the orientation and shape of the organelles on the effective elastic coefficients

The effective elastic coefficients are significantly influenced by the shape and orientation of the organelles. When a compressive displacement is given to the top side of the cell, the effective value of $c_{11,eff}$ is higher for CS-1 compared to CS-2, as depicted in Fig. 11(a). Alterations in the cell's shape and orientation of the organelles resulted in different effective coefficients of the cell structures, particularly the microtubules (Kapat et al. 2020). This phenomenon is also observed when there is an increase in the quantity of microtubules, as it leads to a change in their orientation.

$c_{13,eff}$ exhibits an initial decrease as the number of microtubules is raised from 2 to 6, as depicted in Fig. 12(a). Nevertheless, the orientation of microtubules leads to a significant rise in the effective coefficient, $c_{13,eff}$, when the number of microtubules reaches 10. c_{33} shows a positive correlation with the number of microtubules, increasing as the number of microtubules rises from 2 to 4, as depicted in Fig. 13(a). Nevertheless, it undergoes a sudden decrease once the quantity of microtubules approaches 6. It is demonstrated that the values of $c_{33,eff}$ rise for CS-1 and CS-2, going beyond the number of microtubules 6.

When applying compressive displacement to all sides of the cell except the bottom, $c_{11,eff}$ shows a declining trend as the number of microtubules increases from 2 to 6 for CS-1, as depicted in Fig. 11(b). Nevertheless, it experiences a sharp surge at 10 microtubules, followed by a subsequent decrease. However, unlike CS-1, CS-2 is displaying a distinct pattern. $c_{13,eff}$ is greater for CS-1 when compared to CS-2, as illustrated by Fig. 12(b). The value of $c_{13,eff}$ remains unchanged as the number of microtubules increases from 2 to 6. However, there is a dramatic decrease in $c_{13,eff}$ thereafter, followed by a period of no apparent variation. When subjecting the cell to compressive displacement on all sides, it is evident from Fig. 13(b) that c_{33} increases as the number of microtubules grows for CS-2. However, the effective c_{33} coefficient exhibits minimal fluctuation for CS-1 because of the regular shapes of the organelles.

When the cell is subjected to compressive displacement on all sides (BC-3), it is clear that $c_{11,eff}$ is greater for CS-2 compared to CS-1, particularly for up to 4 microtubules, as shown in Fig. 11(c). The effective value of $c_{11,eff}$ shows a declining pattern in CS-2. This drop is noticed when the number of microtubules is increased from 4 to 6. Moreover, the effective value of $c_{11,eff}$ converges to a comparable value to that of CS-1 when the number of microtubules surpasses 10. $c_{13,eff}$ value is lower for CS-2 compared to CS-1, particularly for microtubule numbers up to 10, as seen in Fig. 12(c). Afterwards, the value of $c_{13,eff}$ approaches convergence with that of CS-1 when considering the number of microtubules beyond 10. $c_{33,eff}$ value for CS-2 is lower than that of CS-1, particularly for 6 microtubules, as shown in Fig. 13(c). However, it approaches the value of CS-1 when considering 6 microtubules. Furthermore, as the number of microtubules reaches 10, there is a noticeable and quick increase in the effective value of $c_{33,eff}$ for CS-1.

Calculating effective elastic coefficients helps to understand the biological cell mechanical behavior depending upon the boundary condition. For instance, cell organelles have a non-uniform distribution, with osteoblasts having cellular elasticity in the cytoplasmic skirt (Charras and Horton 2002; Rigato et al. 2015; Schillers 2019).

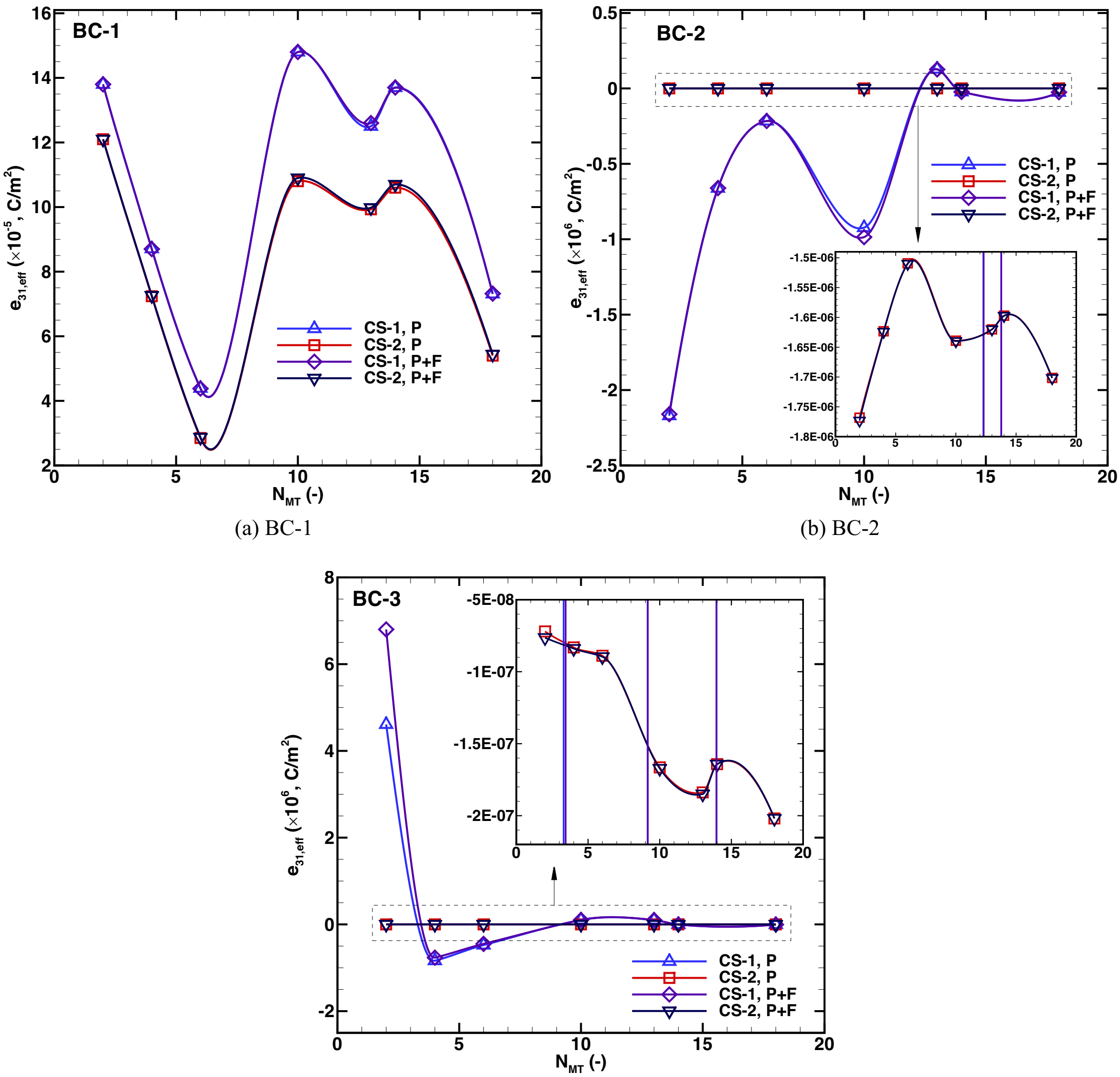
3.4 Determination of the effective piezoelectric coefficients

The presence of piezoelectric inclusions, such as the number of microtubules, significantly impacts the effective elastic moduli of cell structures. The effective piezoelectric coefficients are calculated by dividing the average values of the strain component and the electric flux density vector component, i.e., $e_{31,eff} = \frac{\langle D_3 \rangle}{\epsilon_{11}}$, $e_{33,eff} = \frac{\langle D_3 \rangle}{\epsilon_{33}}$.

3.4.1 Consideration of only the piezoelectric effect and the combination of piezoelectric-flexoelectric effects

Figure 14 displays the effective piezoelectric coefficient ($e_{31,eff}$) within the cell for piezoelectrici effect alone and the combination of piezoelectric and flexoelectric effects, i.e., applying mechanical strain in the x_3-direction causes a piezoelectric reaction, specifically the production of $e_{31,eff}$ in the x_1-direction (Fukada and Yasuda 1957).

For BC-1, it can be observed that $e_{31,eff}$ displays similar trends for both cell structures (CS-1 and CS-2), encompassing both the piezoelectric and flexoelectric phenomena, as depicted in Fig. 14(a). Therefore, the impact of flexoelectricity is insignificant. This phenomenon is caused by the alignment of the microtubules in response to mechanical tension exerted on the top portion of the cell. When it comes to BC-2, the effective value of $e_{31,eff}$ is higher for CS-2 when compared to CS-1, as depicted in Fig. 14(b). The negative values of $e_{31,eff}$ in CS-1 are a result of the compressive stresses in x_1 and x_3 directions caused by the mechanical stress on three sides of the cell when the bottom side is electrically grounded. Therefore, $e_{31,eff}$ is yielding negative values for CS-1. For BC-3, when comparing CS-2 to CS-1, the value of $e_{31,eff}$ shows a trend toward decline with respect to up to four microtubules. This observation relates to the occurrence of piezoelectricity and flexoelectricity. Afterwards, the value of $e_{31,eff}$ shows an increasing pattern and approaches the value of CS-1, as shown in Fig. 14(c).



(a) BC-1 (b) BC-2 (c) BC-3

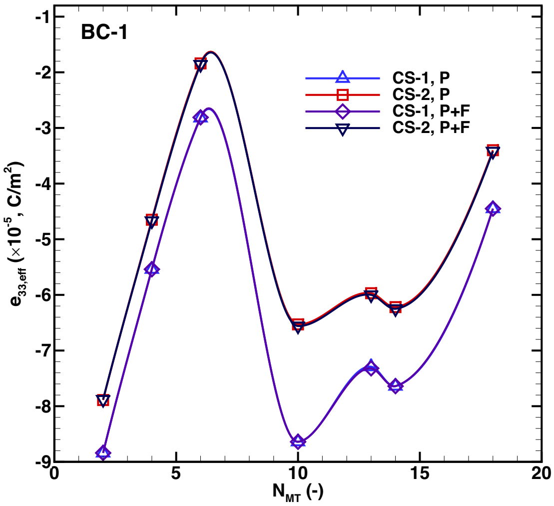
Fig. 14 The effective piezoelectric coefficient ($e_{31,eff}$) of cell structure-1 (CS-1) and cell structure-2 (CS-2) as a function of the number of microtubules considering only the piezoelectric effect (P) and for combined piezoelectric-flexoelectric (P+F) effects are plotted against the compressive displacement at 20 nm for (a) BC-1, (b) BC-2, and (c) BC-3

The effective coefficient $e_{33,eff}$ is calculated for the applied mechanical strain that induces piezoelectric effect in the x_3 -direction, as shown in Fig. 15. It can be observed from Fig. 15(a), (b) that $e_{33,eff}$ is higher for CS-2 than CS-1. This observation holds true for both the piezoelectric and flexoelectric effects in BC-1 and BC-2. Both CS-1 and CS-2 exhibit similar trends for the values of $e_{33,eff}$. In BC-3, the value of $e_{33,eff}$ remains constant for both the piezoelectric and flexoelectric effects until the number of microtubules reaches 10, as depicted in Fig. 15(c), after which it exhibits a significant decrease.

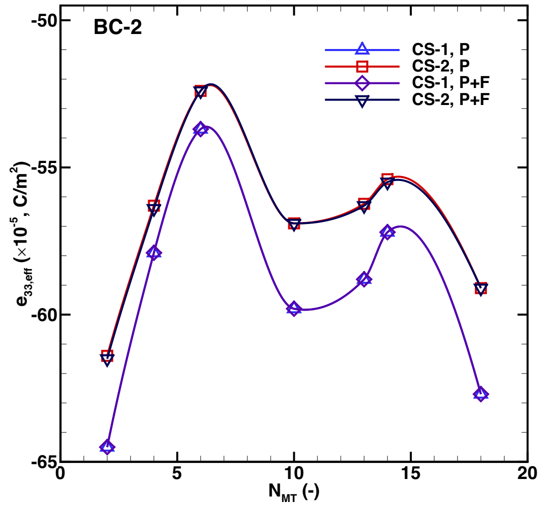
3.4.2 Influence of the organelle's shapes and orientations on the effective piezoelectric coefficients

The effective piezoelectric coefficient, $e_{31,eff}$, has been analyzed under three different applied boundary conditions (BC-1 to BC-3) as a function of the number of microtubules. Despite the symmetry in the shape of the microtubules, the cell structures exhibit different values of $e_{31,eff}$ due to the curved edges at the surfaces.

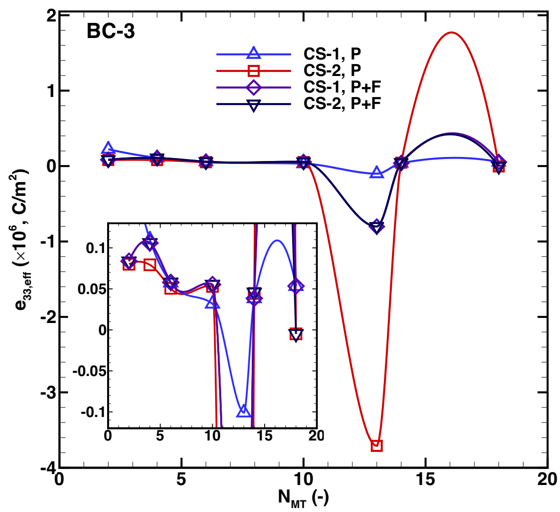
In the case of BC-1, $e_{31,eff}$ initially decreases as the number of microtubules increases from 2 to 6, as shown in Fig. 14(a). However, it subsequently undergoes a significant increase when the number of microtubules reaches 10.



(a) BC-1



(b) BC-2



(c) BC-3

Fig. 15 The effective piezoelectric coefficient ($e_{33,eff}$) of cell structure-1 (CS-1) and cell structure-2 (CS-2) as a function of the number of microtubules considering only the piezoelectric effect (P) and for combined piezoelectric-flexoelectric (P+F) effects are plotted against the compressive displacement at 20 nm for (a) BC-1, (b) BC-2, and (c) BC-3

This behaviour is due to shape variation of microtubules in response to mechanical displacement exerted on the top portion of the cell and geometrical symmetry in the x_3-direction. From the analysis of Fig. 15(a), it is evident that there is an increase in $e_{33,eff}$ for both CS-1 and CS-2 as the number of microtubules is increased from 2 to 6. $e_{33,eff}$ decreases when it transitions from microtubules 6 to 10. Both CS-1 and CS-2 display comparable patterns in terms of the quantity of microtubules for the values of $e_{33,eff}$.

When it comes to BC-2, the value of $e_{31,eff}$ shows no variation regardless of the number of microtubules in cell structure-2, as depicted in Fig. 14(b). Figure 15(b) indicates that there is a rise in $e_{33,eff}$ for both CS-1 and CS-2 when the number of microtubules is raised from 2 to 6 for BC-2. The value of $e_{31,eff}$ shows an increasing pattern and approaches the value of CS-1 when the number of microtubules exceeds 10, as shown in Fig. 14(c). This behaviour has been caused by the orientation of the microtubules. Moreover, the stress is exerted uniformly on all sides of the cell, leading to no strain gradient. $e_{33,eff}$ consistent for the number of microtubules increases from 2 to 10; beyond that, it experiences a notable reduction, as shown in Fig. 15(c).

In summary, a linearly coupled electromechanical model has been developed to quantify cellular mechanics under external mechanical displacement, considering piezoelectric and nonlocal flexoelectric effects. The present model assumes that living cells are linear elastic, which provides the first step in the analysis (Kollmannsberger and Fabry 2011; Garcia and Garcia 2018b). Living cells are viscoelastic, and incorporating dynamic analysis into a three-dimensional study could provide more realistic responses with variable shapes of the organelles and cell structures (Chaudhuri et al. 2020). Another major limitation of the present work is considering the biological cell in two dimensions and performs only the steady-state analysis. Incorporating dynamic analysis into the three-dimensional model is expected to enhance the precision and accuracy of cellular mechanical responses. This advancement holds potential applications in various medical technology fields, including sensing and energy-harvesting device fabrication. Future extensions could also include cellular tensegrity models, which account for individual components of the cytoskeleton using discrete beam or truss elements (Ingber et al. 2014; Sun et al. 2023). Clearly, in realistic scenarios, the electro-elastic properties of biological cells are not uniform and exhibit different qualities in all directions. To capture the intricacies of these phenomena, more sophisticated multiscale modelling methodologies would need to be developed (Ambrosi et al. 2019; Schofield et al. 2020). Despite limitations, the developed coupled electromechanical model is an important initial model that incorporates piezoelectric and flexoelectric effects within the biological cell. Comparative analysis of the predicted outcomes could help understand the complex electromechanical response of biological cells and assist in the design of electromechanical devices for various medical applications.

4 Conclusions

The presented study has explored a two-dimensional conceptual framework of the biological cell for two different structures, encompassing many essential organelles that experience distinct mechanical loading conditions originating from the external environment. The impact of incorporating flexoelectric coupling into the electromechanical model is significantly more prominent than that of solely introducing piezoelectric coupling. The cell structures used in the present study are classified into two categories: regular and arbitrary shapes. Understanding cell mechanics is highly dependent on these structures and the boundary conditions since they play a significant role in influencing cell behaviour and associated stimulation. A parametric analysis revealed that cell structures with regular shapes of the organelles generate a higher maximum electric potential when combined with piezoelectric and flexoelectric effects; this indicates that there are more microtubules in the cellular structure. The mechanical and piezoelectric responses of the cell under three distinct boundary conditions have been elucidated with the help of the effective elastic and piezoelectric coefficients. Cell structures show that the presence of piezoelectric inclusions, specifically microtubules, significantly affects effective elastic moduli, leading to increased effective moduli. The number of microtubules significantly influences the effective elastic coefficients and piezoelectric coefficients, both with and without considering the piezoelectric properties, due to the dependence of the behaviour of cells on their structure, the orientation of the microtubules, and the direction of mechanical stress. Microtubule number, orientation, and loading conditions significantly impact cell mechanics, predicting increased electromechanical effects. The model developed and the findings presented in this study will provide valuable insights into the mechanics of living cells that are not easy to understand through experiments. The model developed in this work can be extended to a three-dimensional space to accurately represent the dynamic behaviour of biological cells under various boundary conditions. The proposed methodology is also useful in quantifying the influence of linear piezoelectric and nonlocal phenomena when external mechanical stimuli are applied. This would facilitate obtaining vital knowledge necessary for understanding the mechanics at the cellular level.

These findings have practical applications in the development of biocompatible nano-biosensors, drug delivery systems, noninvasive diagnostic techniques, and therapeutic approaches where the properties of biological cells need to be better understood.

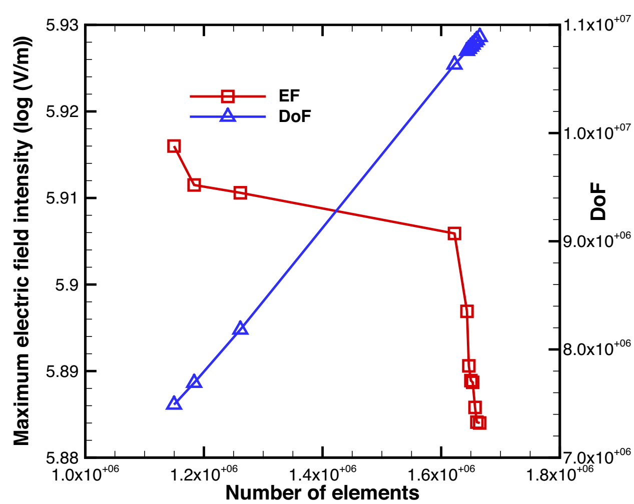


Fig. 16 The relationship between the total number of elements and the electric field intensity, as well as the degrees of freedom (DoF)

Table 3 Details of the mesh independence analysis

S.No	Number of elements	Degrees of freedom	Maximum electric field intensity ($\log(V/m)$)	Relative % error
1	1,149,649	7,489,668	5.9160	–
2	1,183,564	7,691,148	5.9115	0.0761
3	1,261,426	8,183,232	5.9106	0.0152
4	1,622,430	10,632,984	5.9059	0.0795
5	1,643,463	10,759,200	5.8969	0.1526
6	1,646,619	10,778,136	5.8906	0.1069
7	1,650,073	10,798,860	5.8889	0.0288
8	**1,653,103**	**10,817,040**	**5.8887**	**0.0033**
9	1,657,046	10,840,704	5.8858	0.0492
10	1,660,121	10,859,160	5.8841	0.0288
11	1,664,968	10,888,236	5.8840	0.0016

Appendix A. Mesh Independence analysis

The precision of numerical solutions is contingent upon the specific mesh employed to discretize the governing equations and boundary conditions. As the mesh size decreases and tends to zero within its optimal limits, the accuracy of the solutions improves, but the computational cost increases. The degree of freedom needed for model computation significantly impacts time and memory use. It is typically advantageous to estimate the degrees of freedom by considering the number of mesh elements in the model. Hence, a mesh independence analysis has been performed on CS-2 over the entire computational domain using a non-uniform triangular mesh to evaluate the trade-off between the accuracy of the solution and computational cost.

Table 3 presents pertinent data regarding the quantity of elements, degrees of freedom, and the maximum electric field intensity within the cell. The present study has noted an increase in the computational cost as a result of the irregular structure of the microtubules and shapes of the mitochondria. This ultimately leads to an increase in both the number of elements and the degrees of freedom, as depicted in Fig. 16. The trend observed is that the maximum electric field intensity value exhibits an increasing pattern as the number of elements increases up to 1650073. After that, the changes in the solution are negligible in terms of the maximum electric field intensity and relative percentage error. For instance, at 1,653,103 elements, and the maximum electric field intensity value is 5.887 V/m, with a relative percentage error of 0.0033. Therefore, in order to optimize computational efforts and cost, the total number of elements selected and all subsequent results presented in this work have been given for 1,653,103 elements and the degrees of freedom, 10817040. The mesh independence analysis for CS-1 has been conducted in a separate study (Singh et al. 2020).

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